

Learning and Behavior

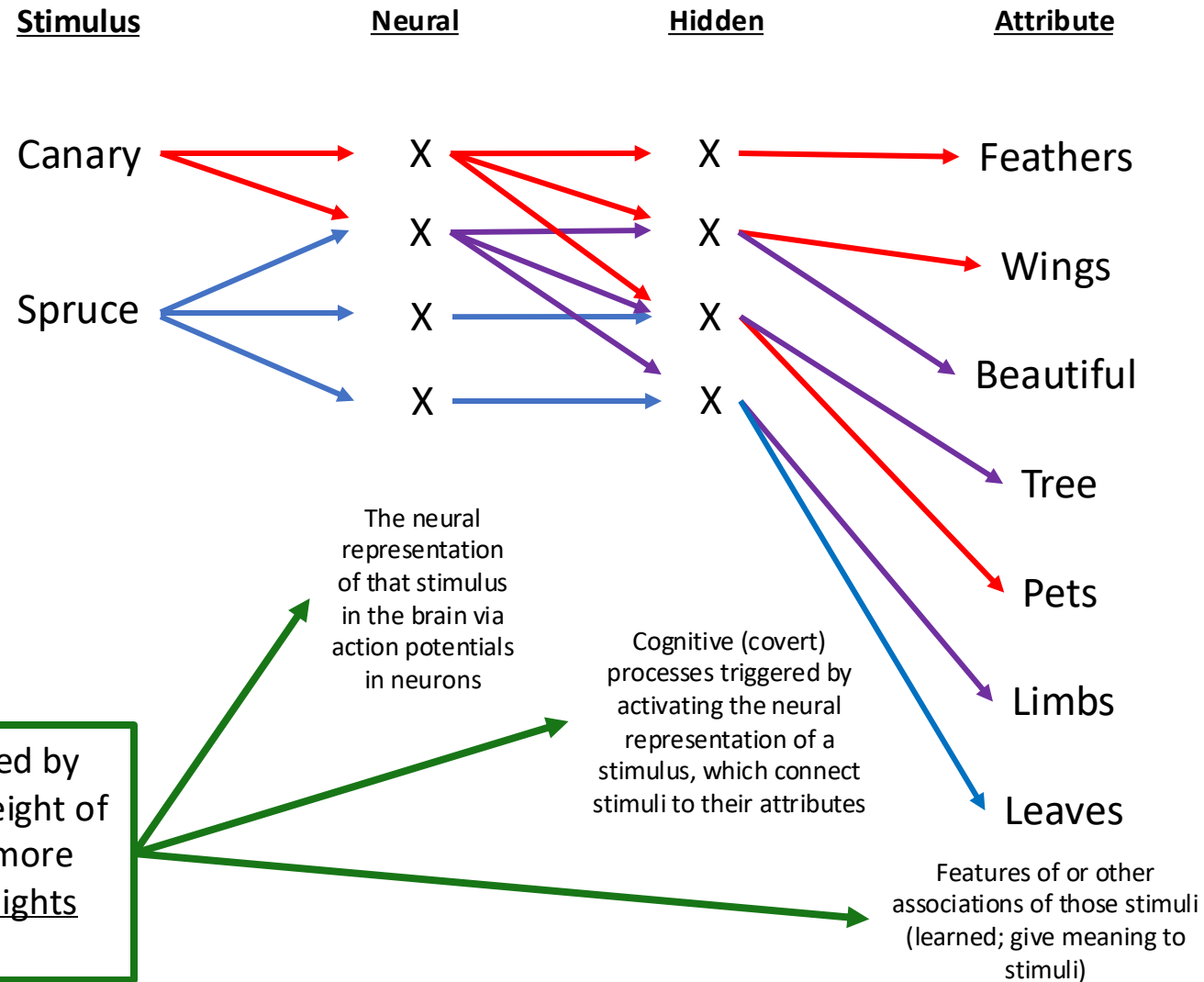
PSY/NSC 3320

Jean P. Bodet III, Ph.D.

Baylor University

THE ROAD SO FAR

The Connectionist Model



All of these relationships are mediated by **weighted connections** – stronger weight of connection between two concepts, more likely that one triggers the other; weights can change based on experience

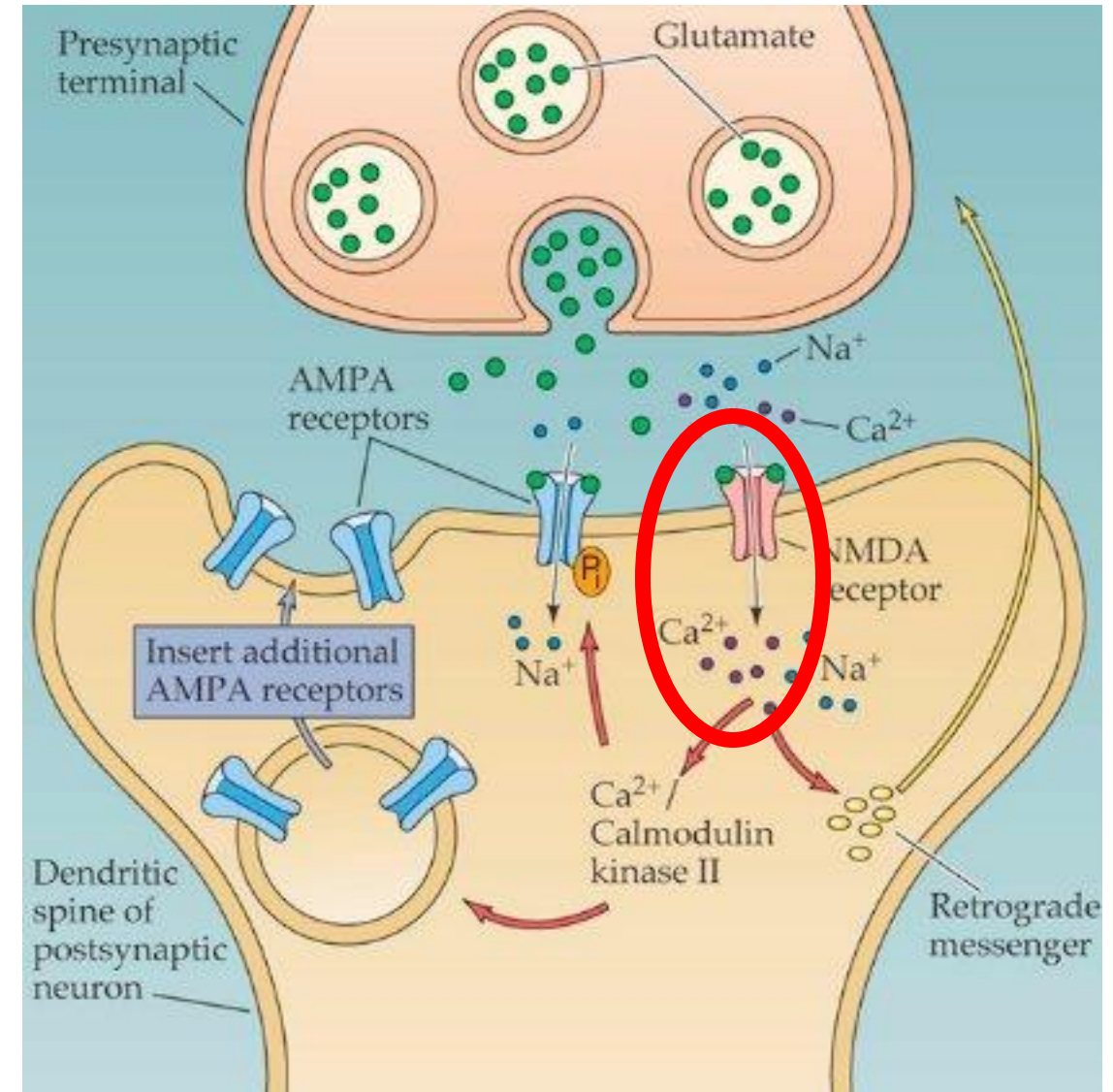
Glutamate Receptor Types

AMPA

- Open when Glutamate from presynaptic binds to it
- Allow **Na⁺** in
- **Depolarizes (*excites*) postsynaptic neuron**

NMDA

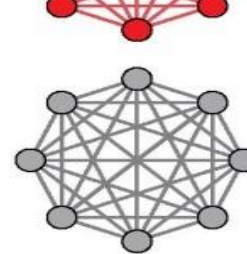
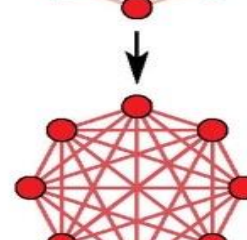
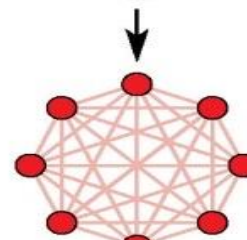
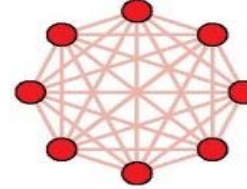
- Open when Glutamate from presynaptic binds to it AND postsynaptic is firing
- Allow **Ca⁺⁺** in
- **Triggers adaptation of postsynaptic neuron by adding new AMPA receptors at that synapse**



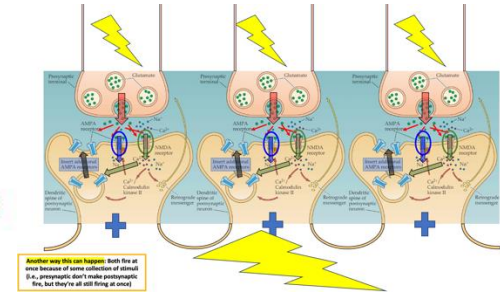
First presentation of stimulus → Neural representation



External stimulus



LONG-TERM POTENTIATION AT GLUTAMATERGIC SYNAPSES OF THE NEURONS INVOLVED IN PROCESSING THIS STIMULUS THROUGHOUT THE BRAIN

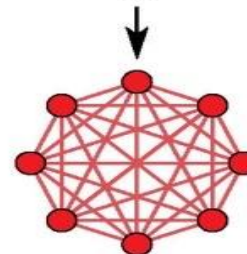
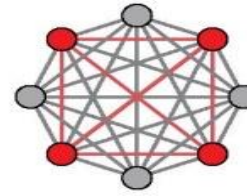


Fire together → Wire together

Wires (strengthened connections) remain after firing stops

TIME PASSES

Later, whole or partial stimulus



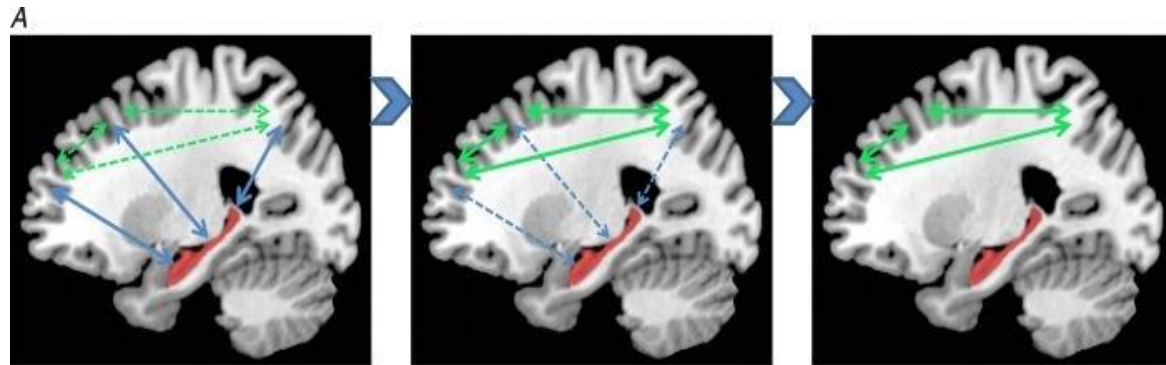
Reactivates whole cell assembly

RECAP

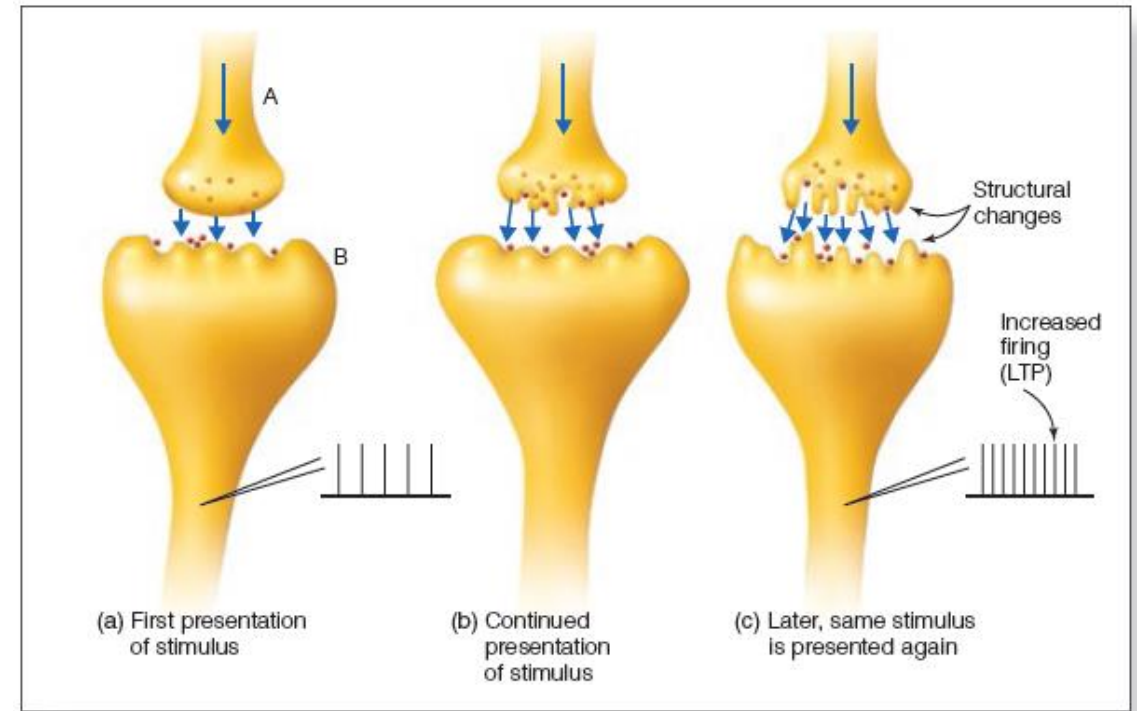
NOW

The Neuroscience of Learning: Systems Consolidation

Systems Consolidation

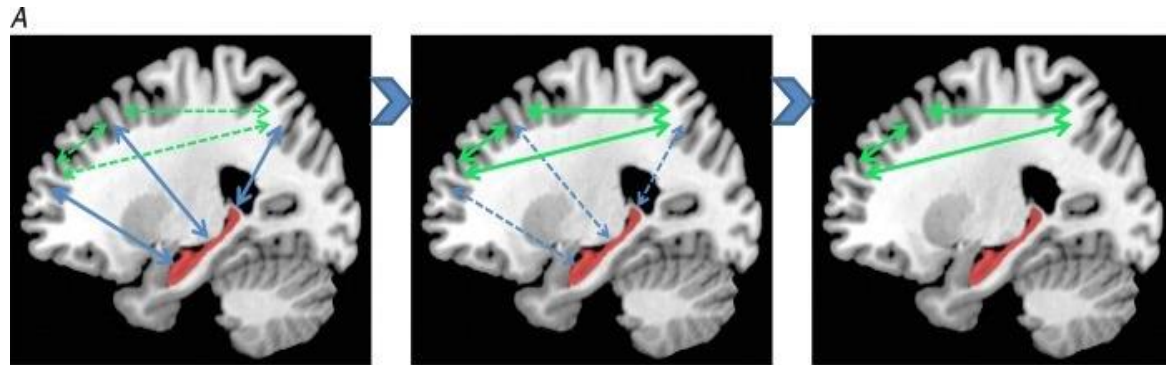


Synaptic Consolidation



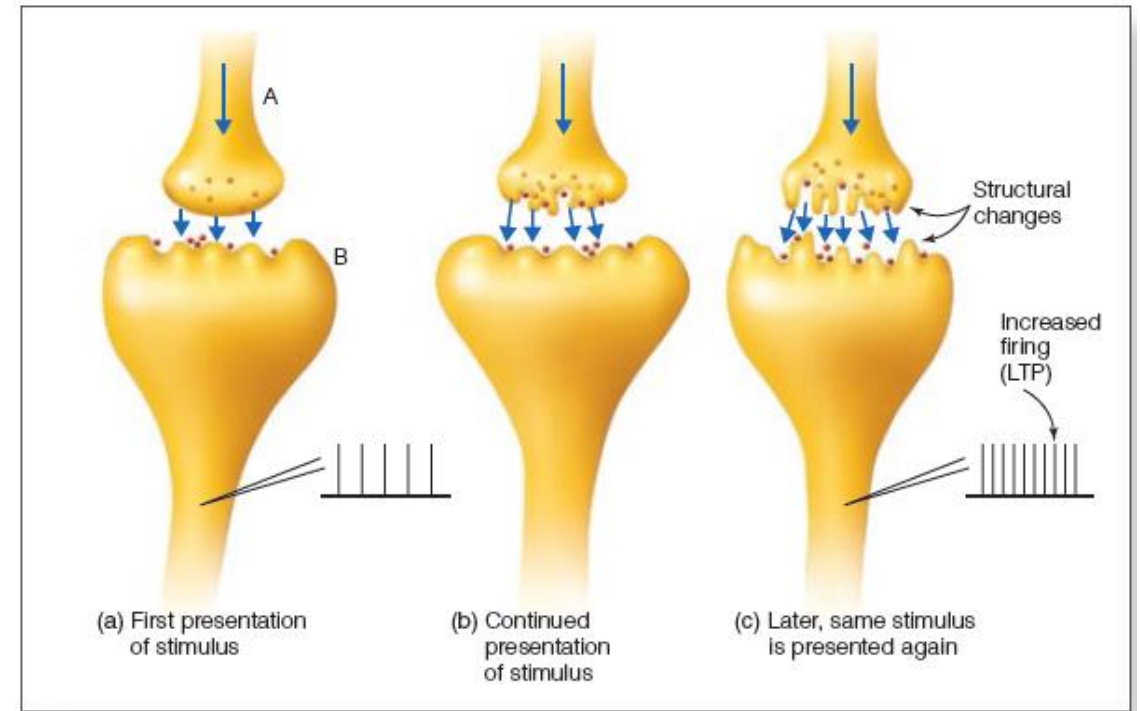
*How this works depends
on the type of learning*

Systems Consolidation



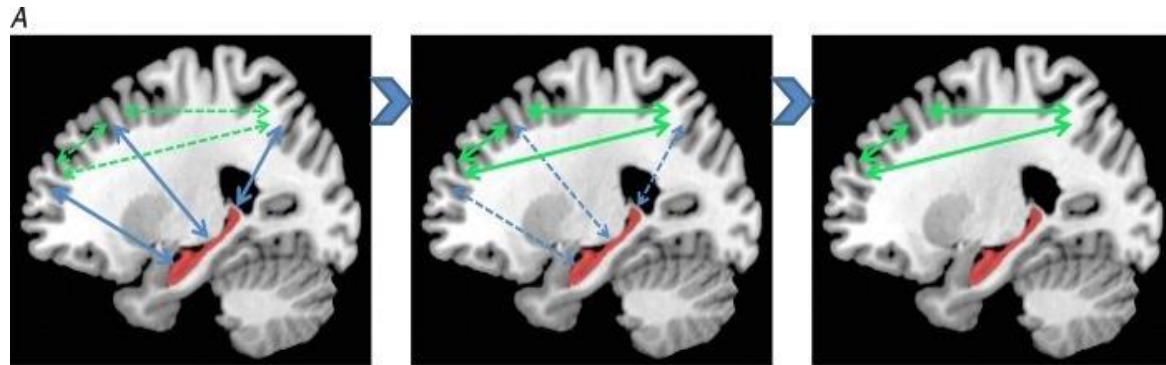
*This works the same way no
matter the type of learning*

Synaptic Consolidation



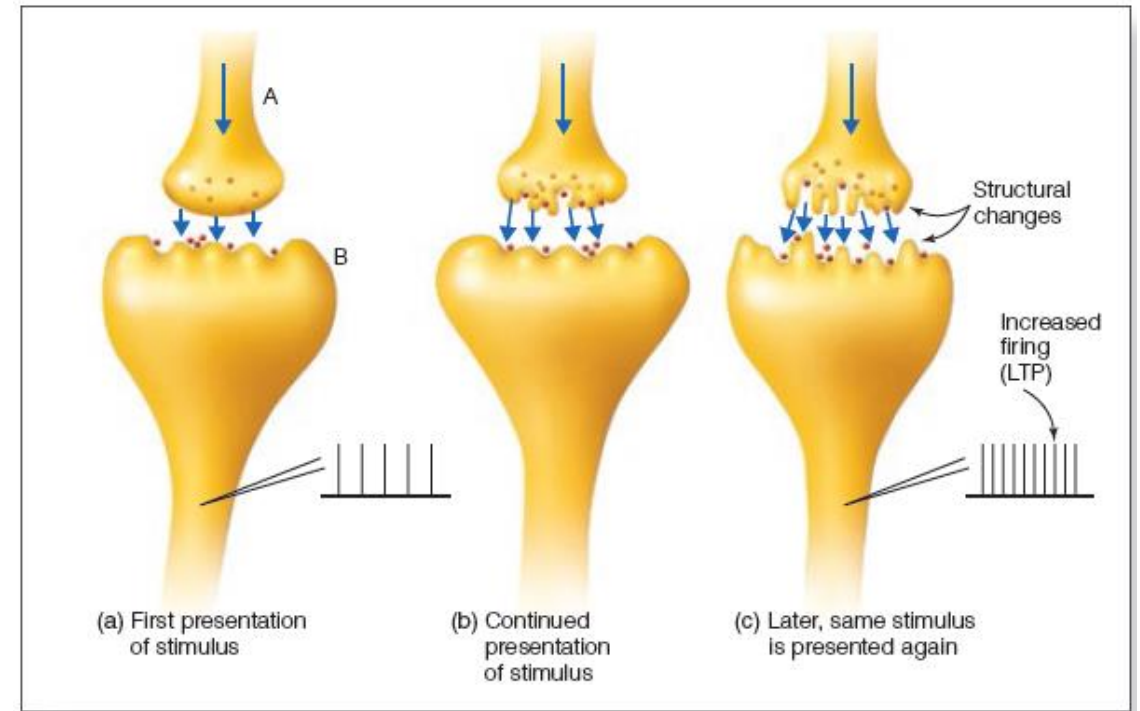
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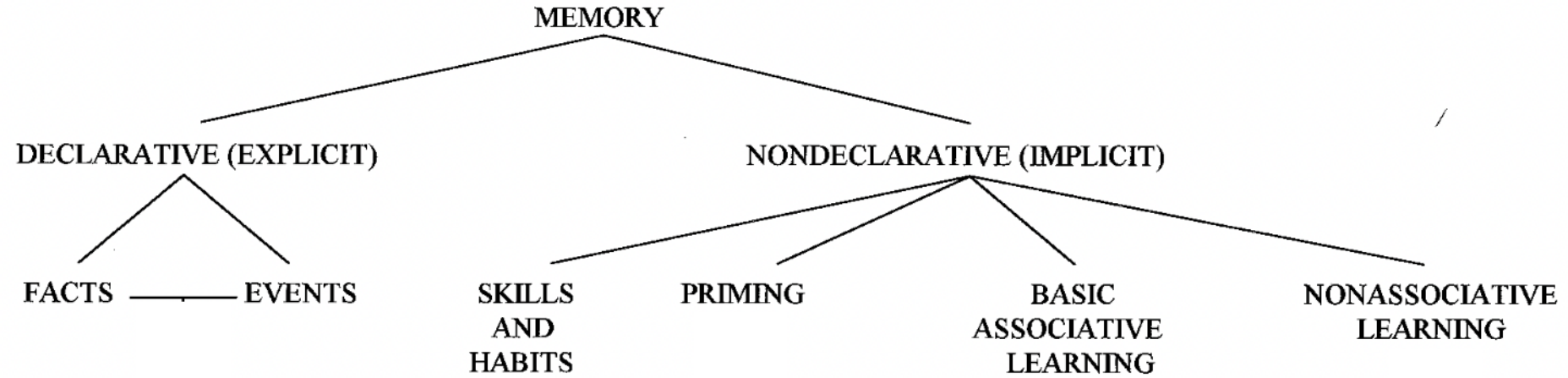
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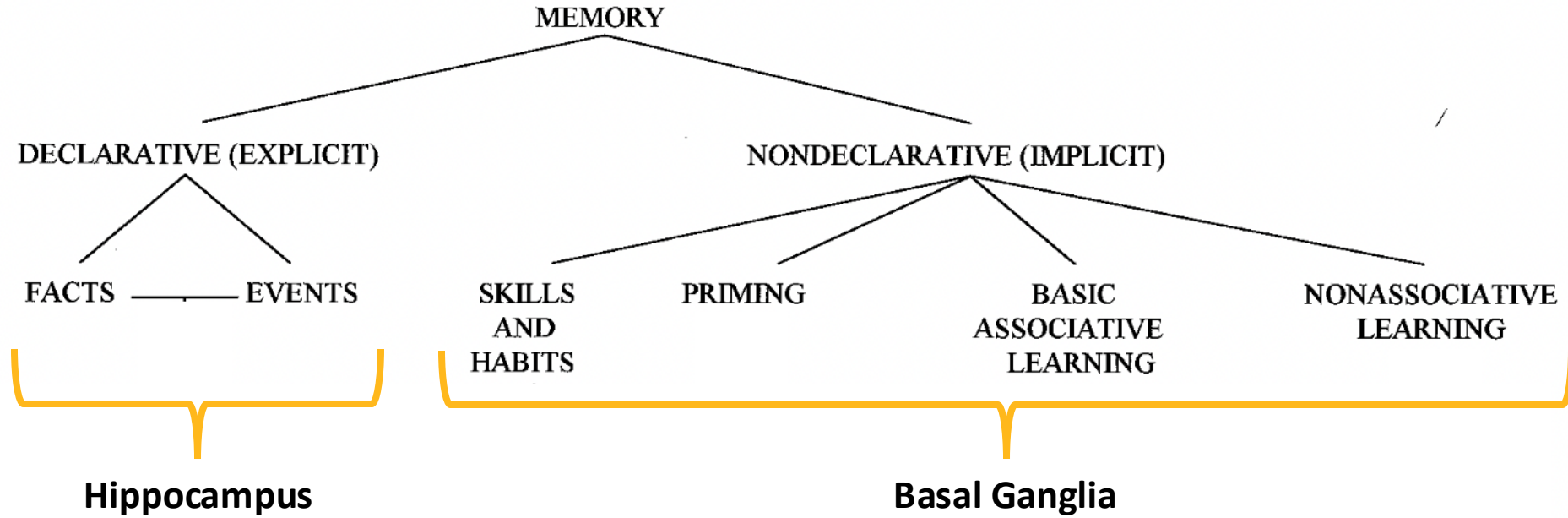


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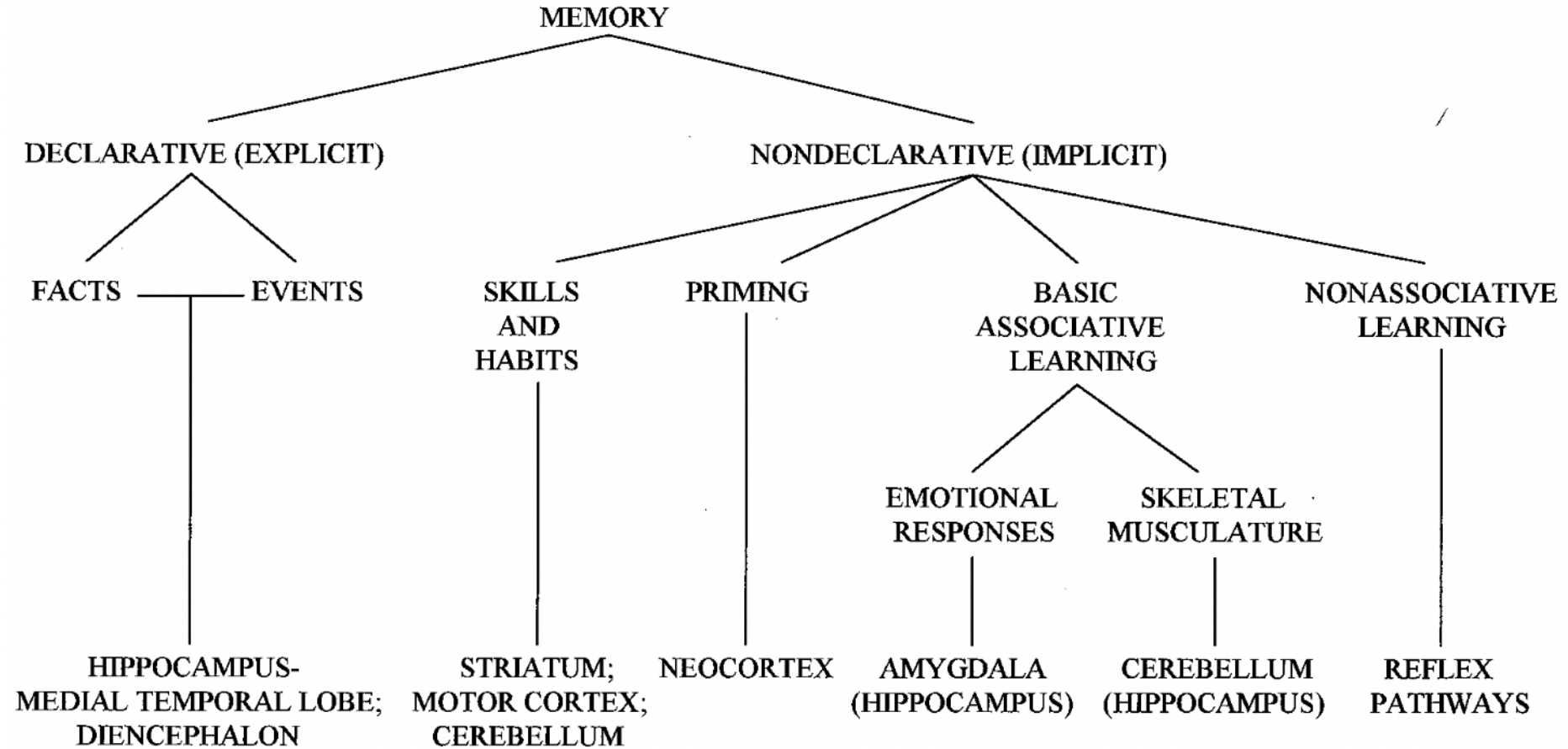
Synaptic Consolidation



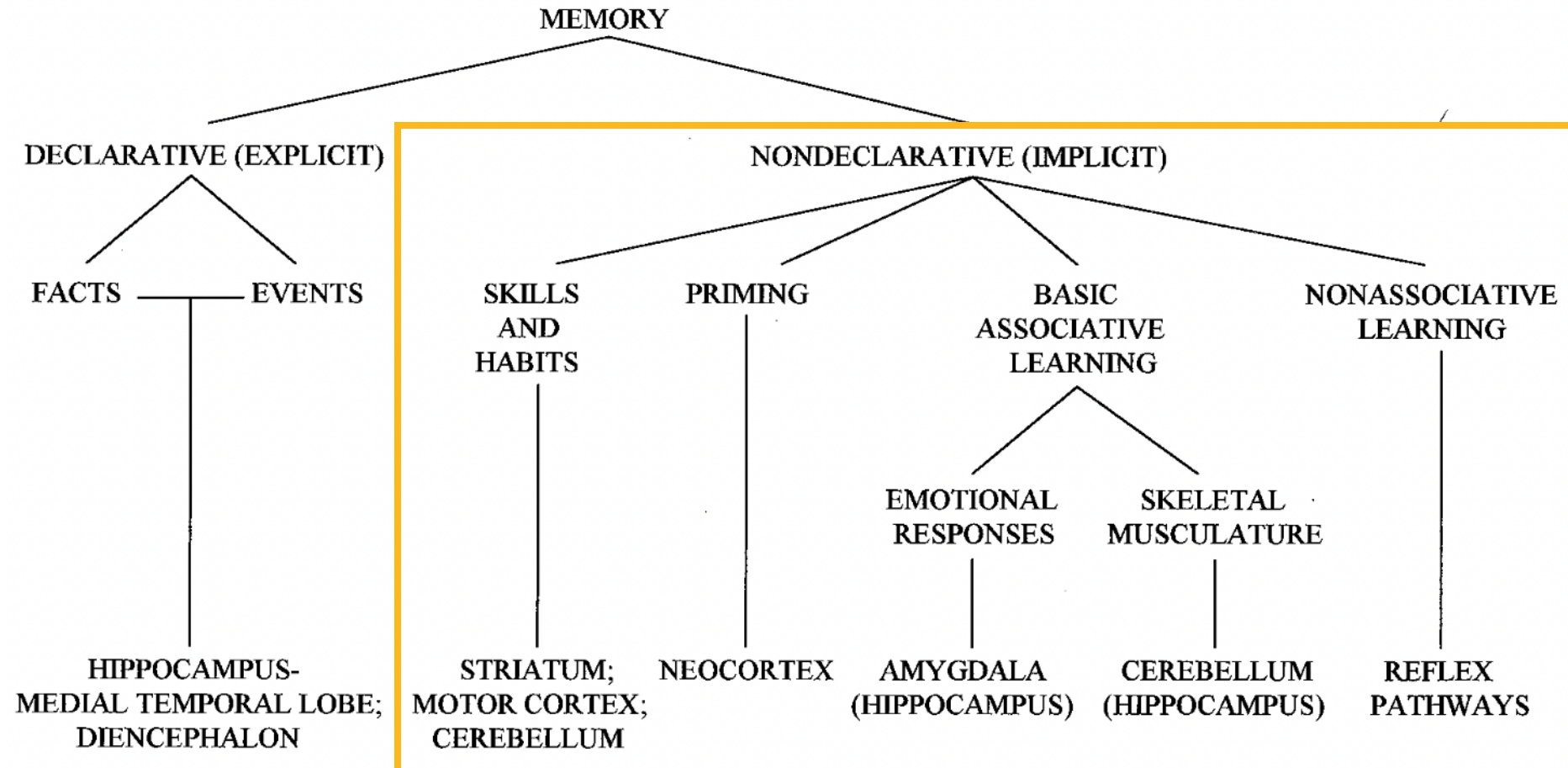




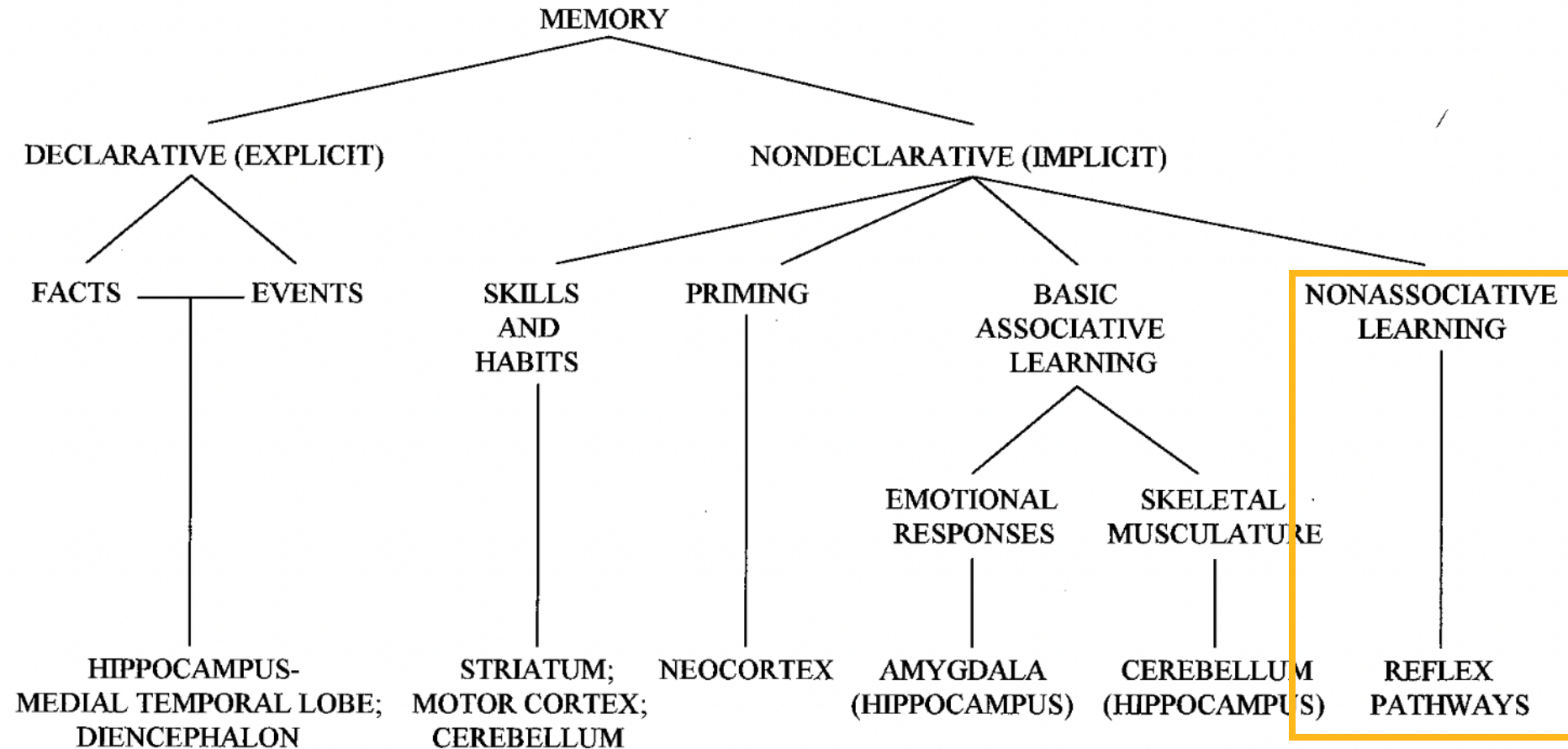
"Two-System" approaches to neuro of learning



"Multiple-System" approaches to neuro of learning

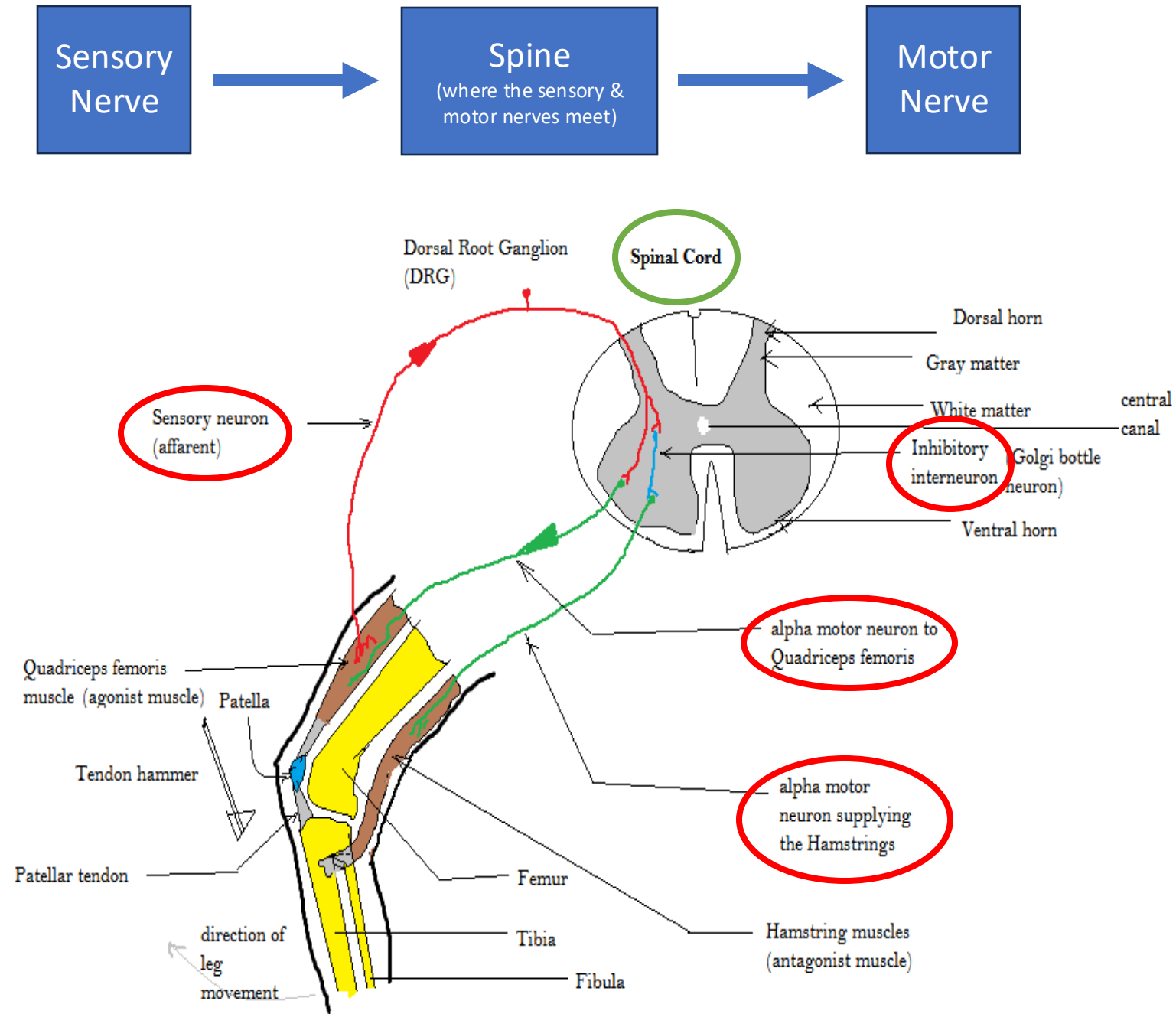


The Neuroscience of Learning: Systems Consolidation Implicit Learning



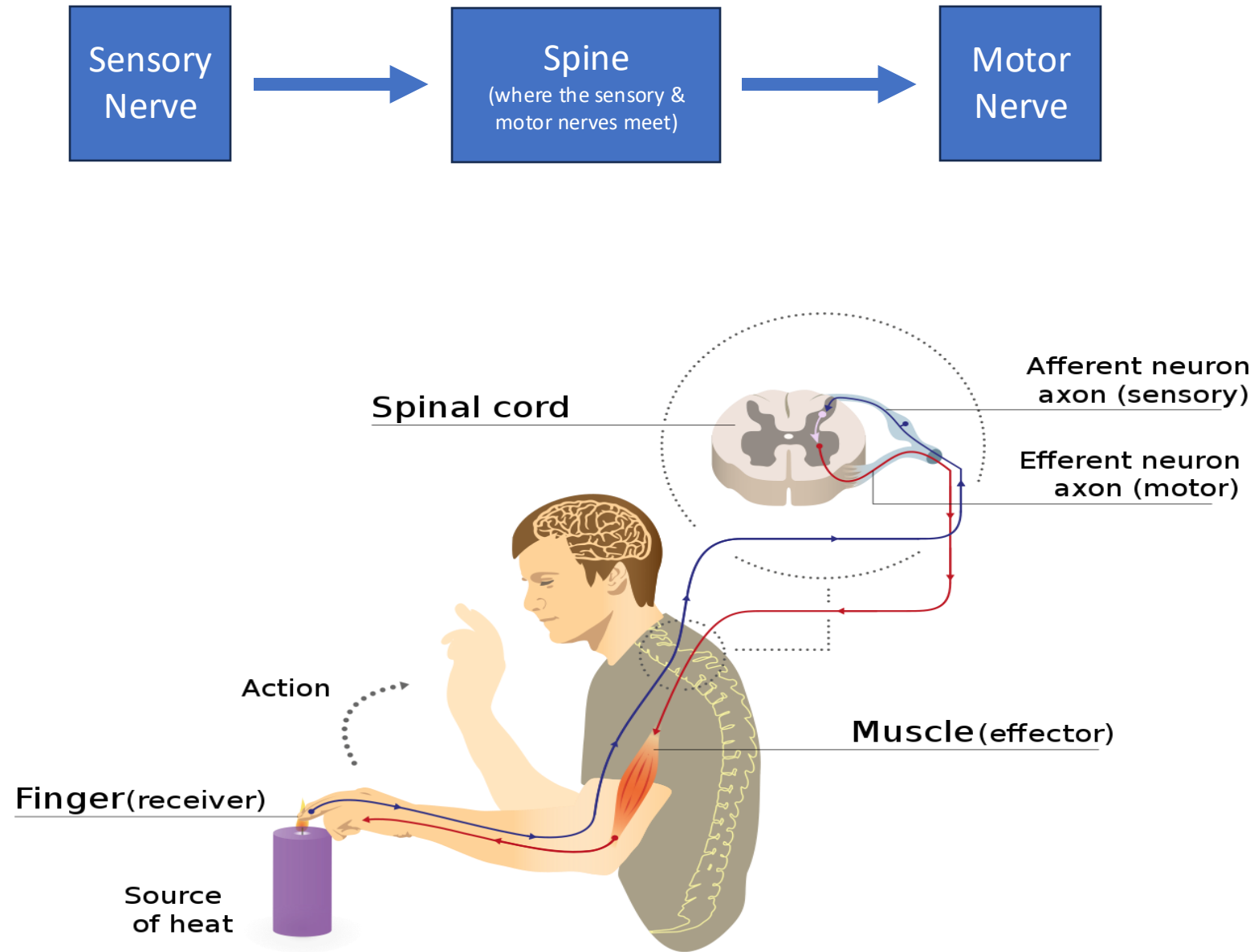
Reflexes – automatic response to a stimulus

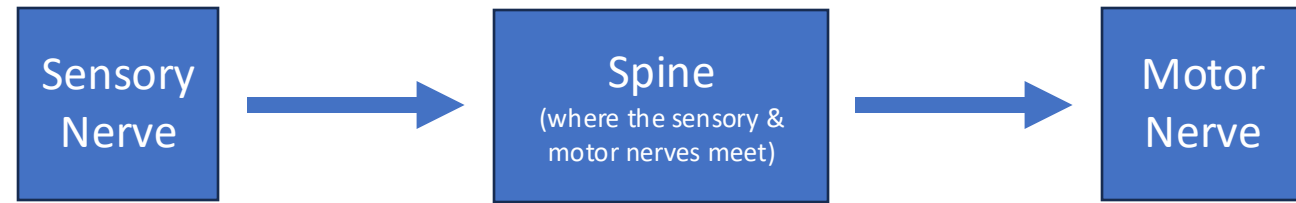
- Reflex Arcs – *spine* – simple, in-born reflexes



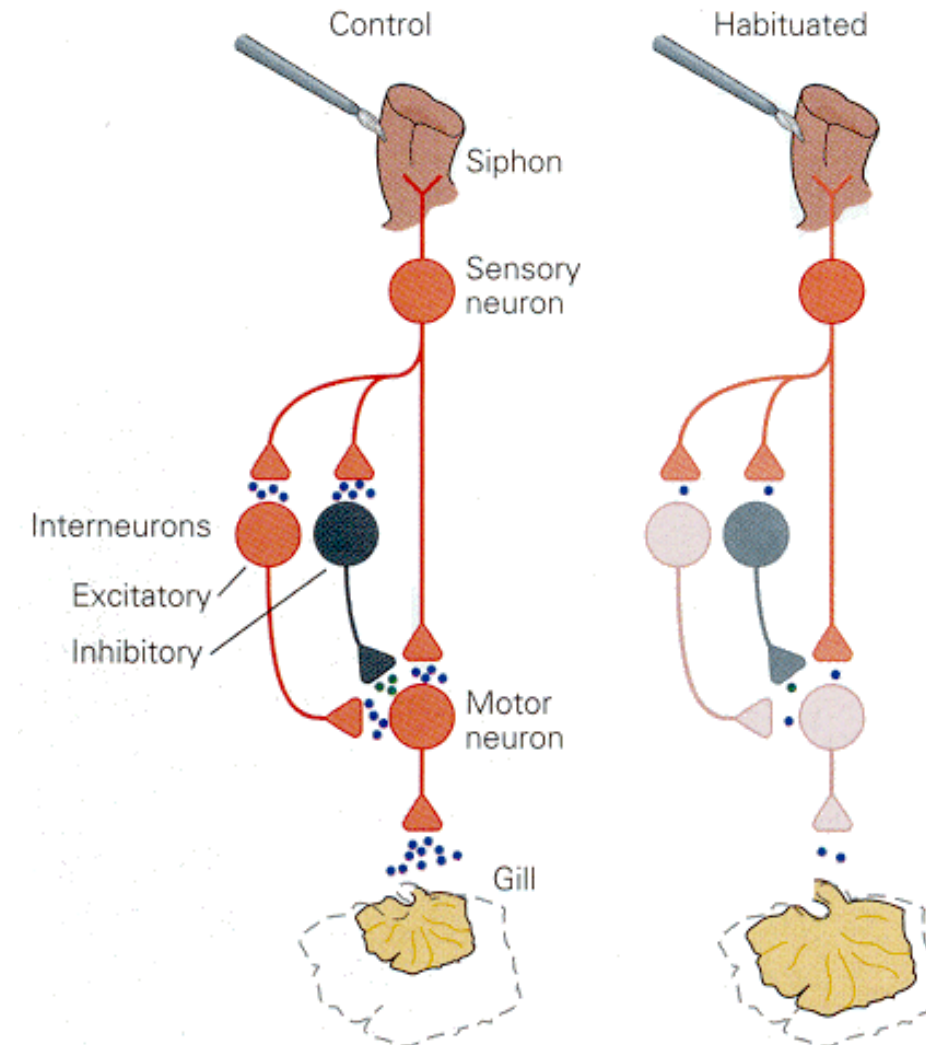
Reflexes – automatic response to a stimulus

- Reflex Arcs – *spine* – simple, in-born reflexes
- Flexion Response – *spine* – auto-recoil from hot/sharp object (*step on nail, don't 'feel' till after*)





Repeated exposure → Change in influence of sensory nerve on motor nerve(s)



Reflexes – automatic response to a stimulus

- Reflex Arcs
- Flexion Response

Two types of **Non-Associative Learning**:

- Habituation
- Sensitization

*The biology of Non-Associative Learning (shown here) is **not** the same as the biology of LTP*

Reflexes – automatic response to a stimulus

- Reflex Arcs
- Flexion Response

Two types of **Non-Associative Learning**:

- Habituation
- Sensitization

Sensory Nerve



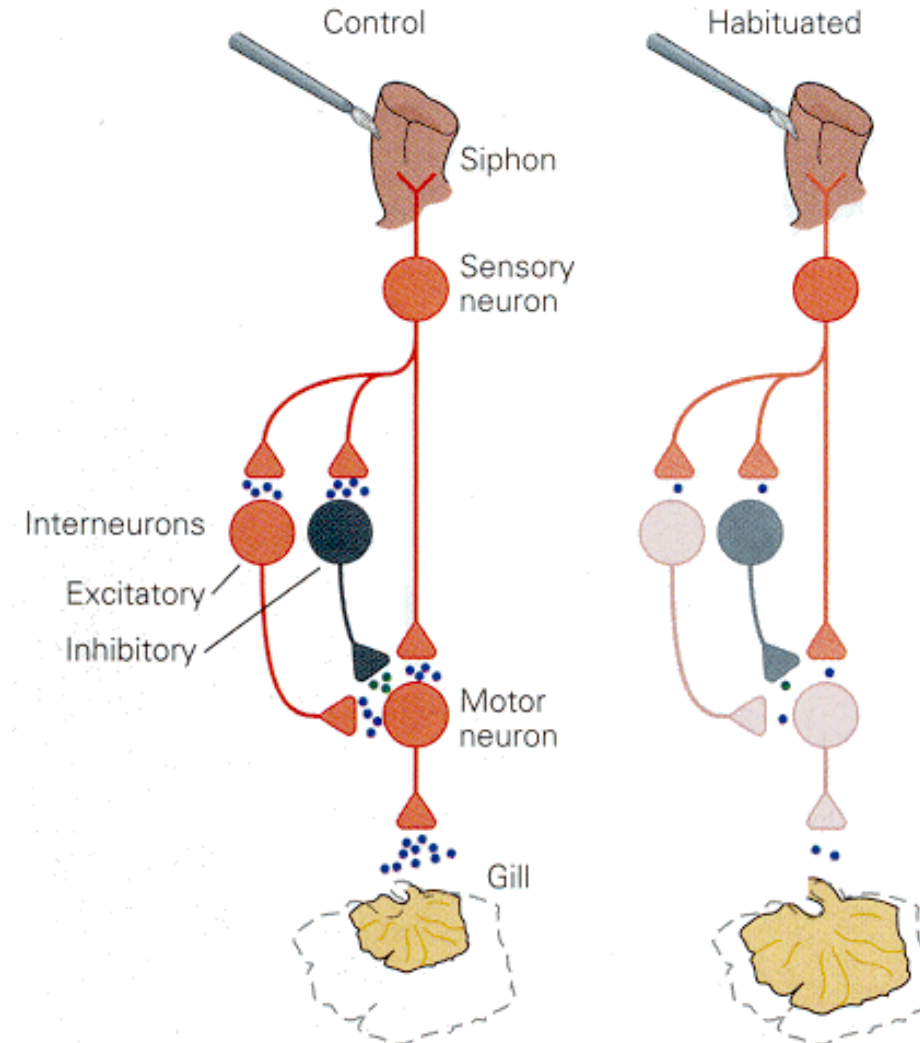
Spine

(where the sensory & motor nerves meet)



Motor Nerve

Repeated exposure → Change in influence of sensory nerve on motor nerve(s)

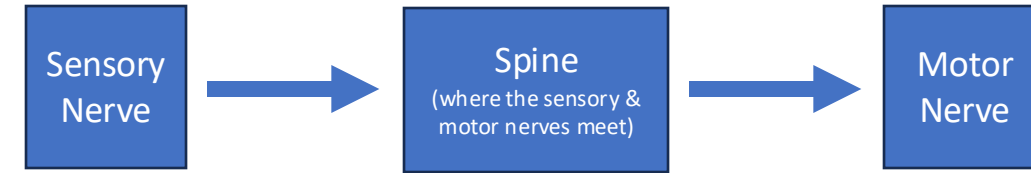
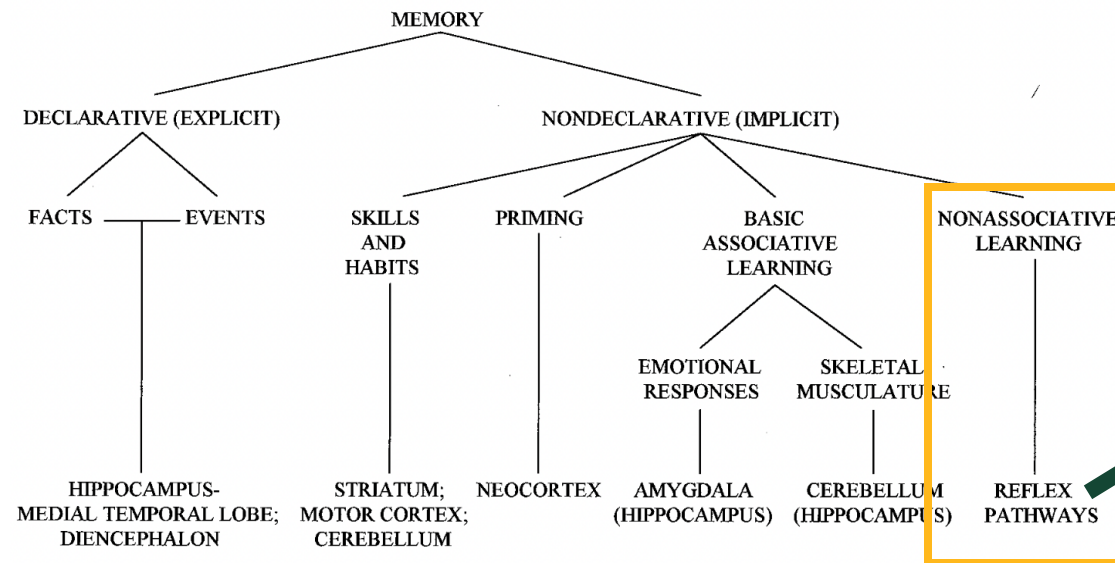


Reflexes – automatic response to a stimulus

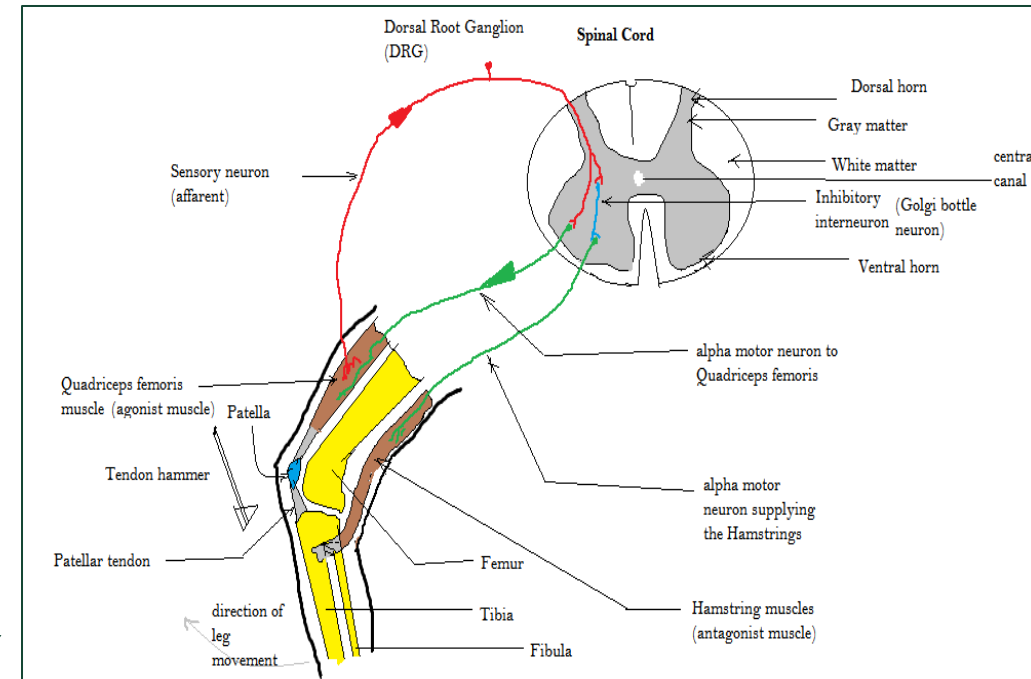
- Reflex Arcs
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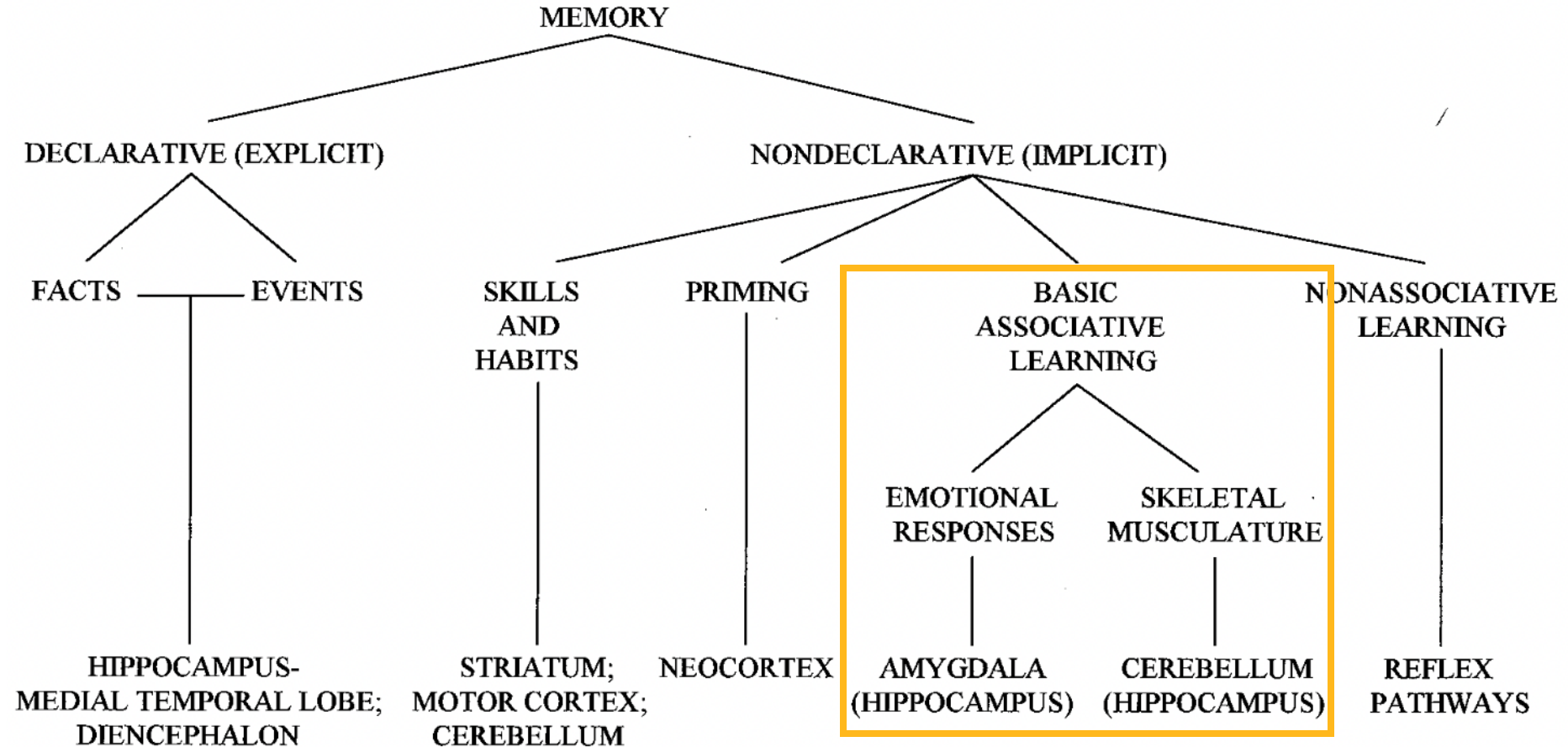
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Repeated exposure → Change in influence of sensory nerve on motor nerve(s)

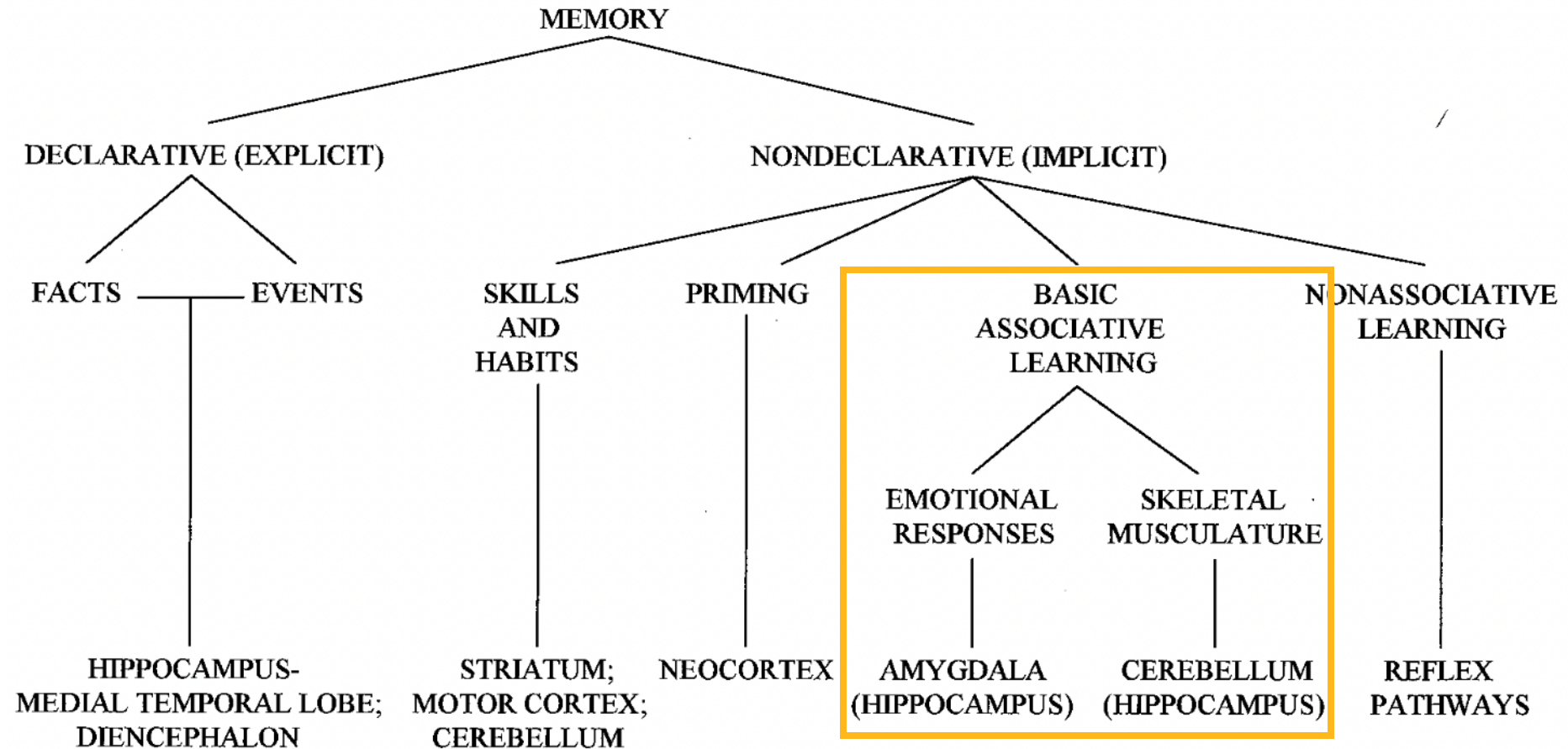




Operant (choice behaviors)

Classical (elicited behaviors)

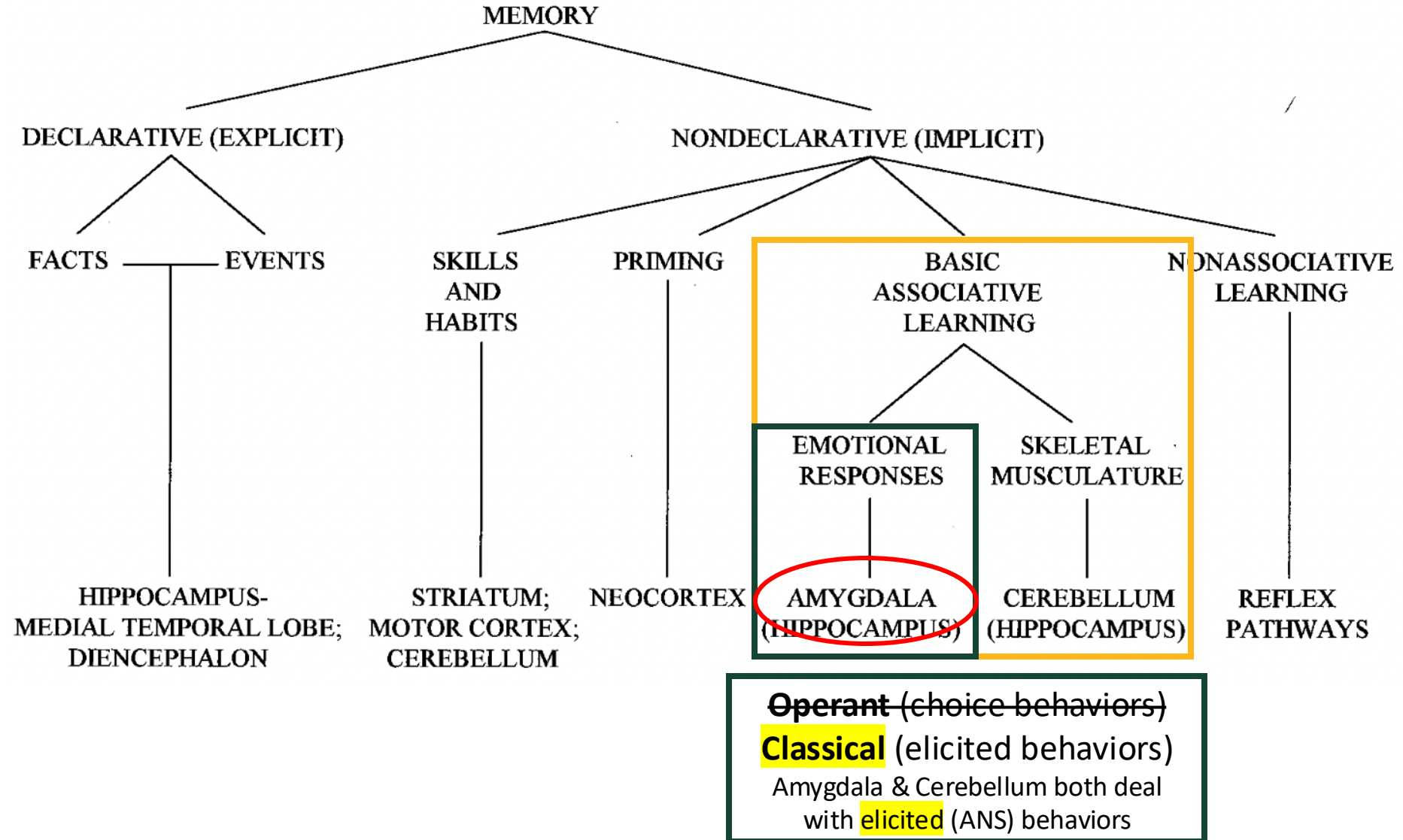
Amygdala & Cerebellum both deal with elicited (ANS) behaviors



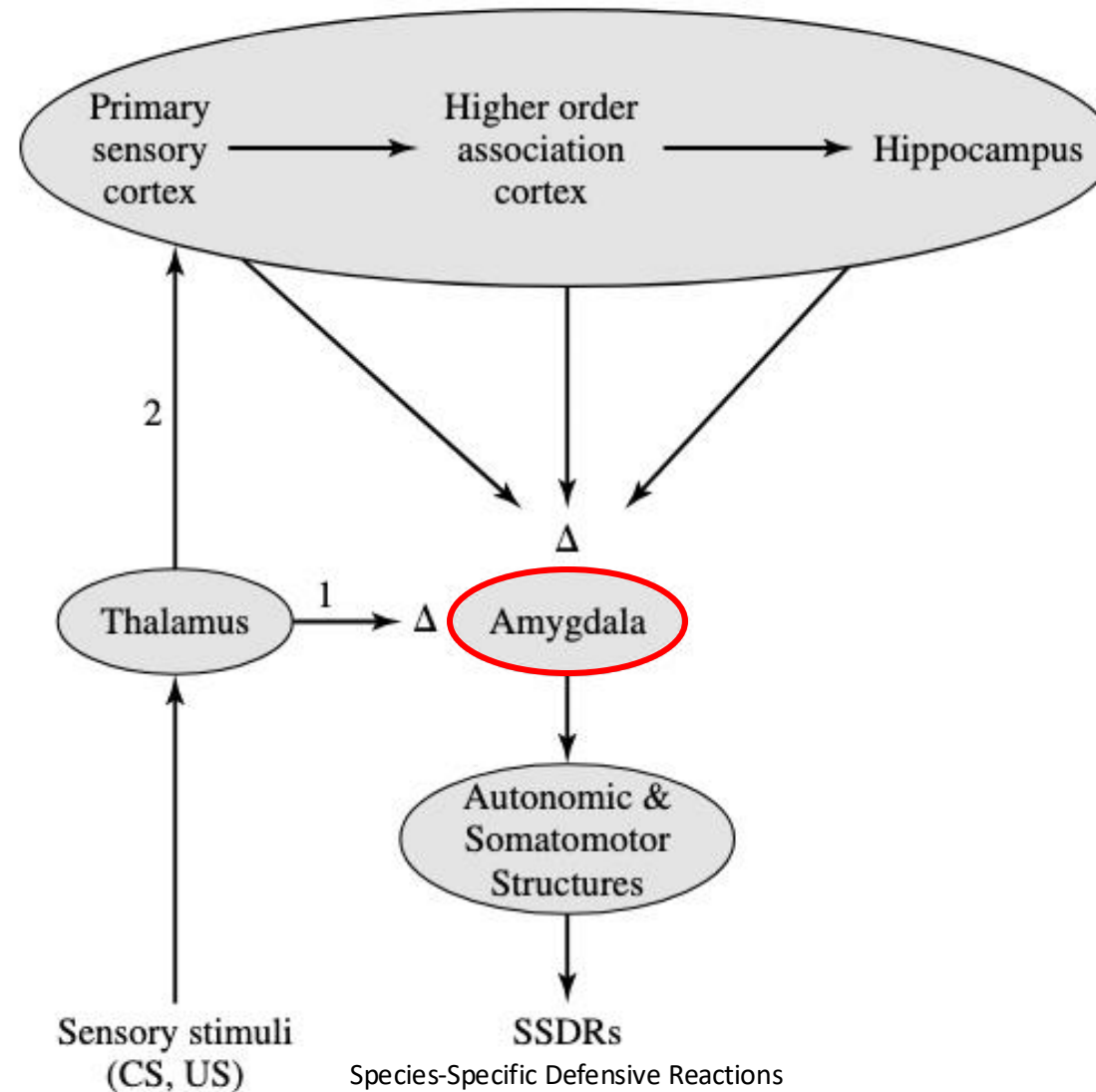
~~Operant~~ (choice behaviors)

Classical (elicited behaviors)

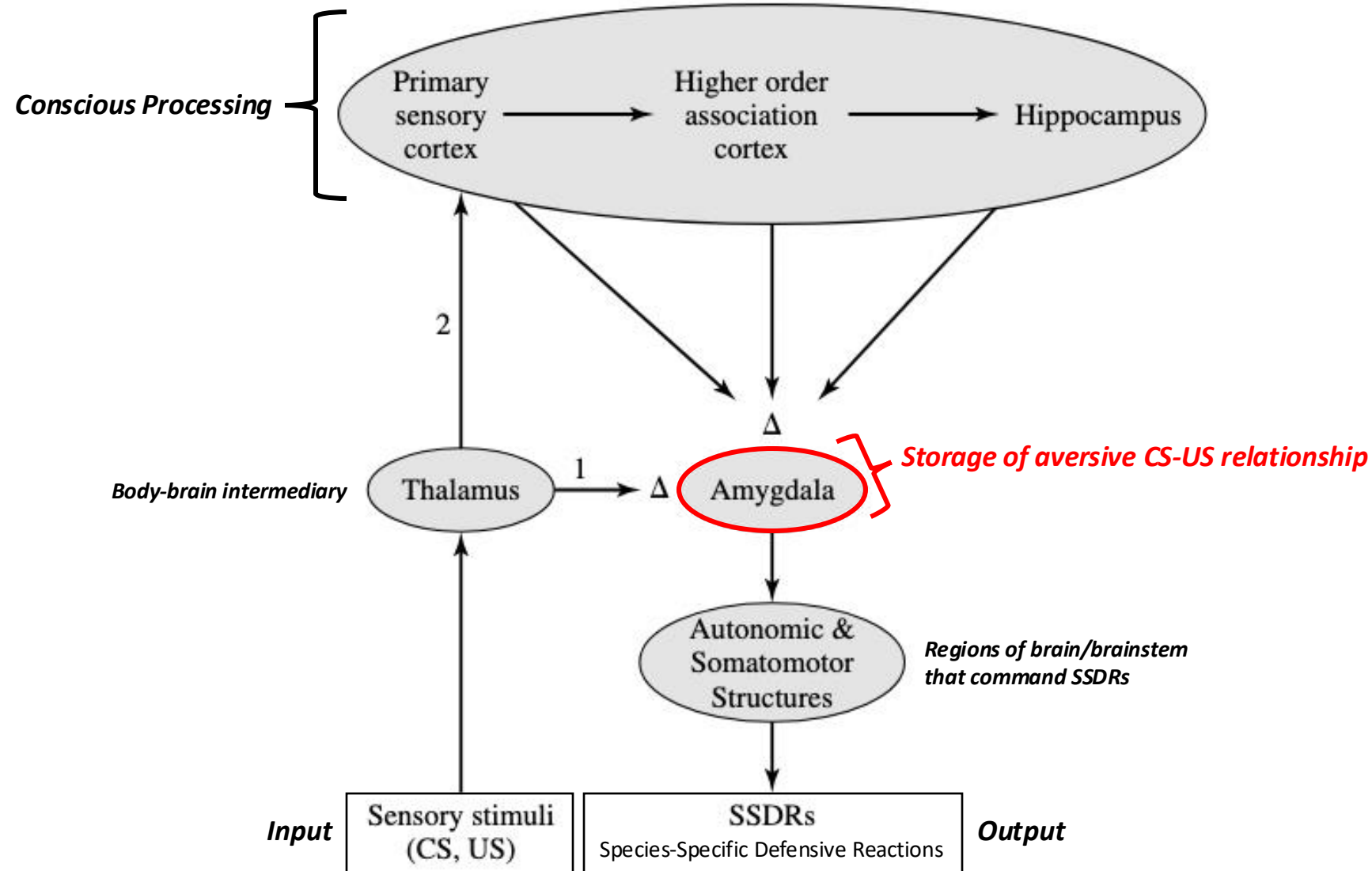
Amygdala & Cerebellum both deal with elicited (ANS) behaviors



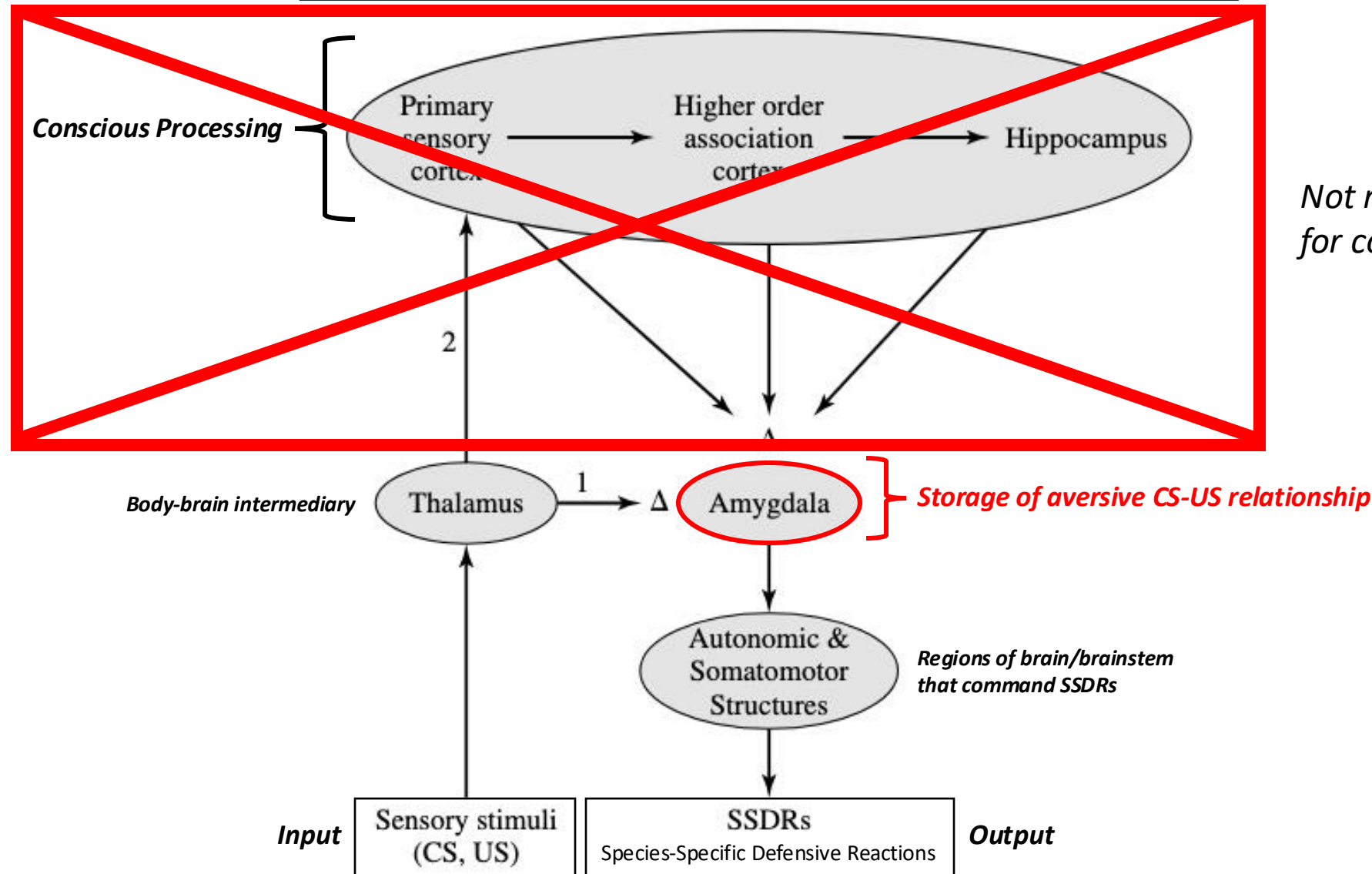
A simplified model of the neuroscience of aversive conditioning



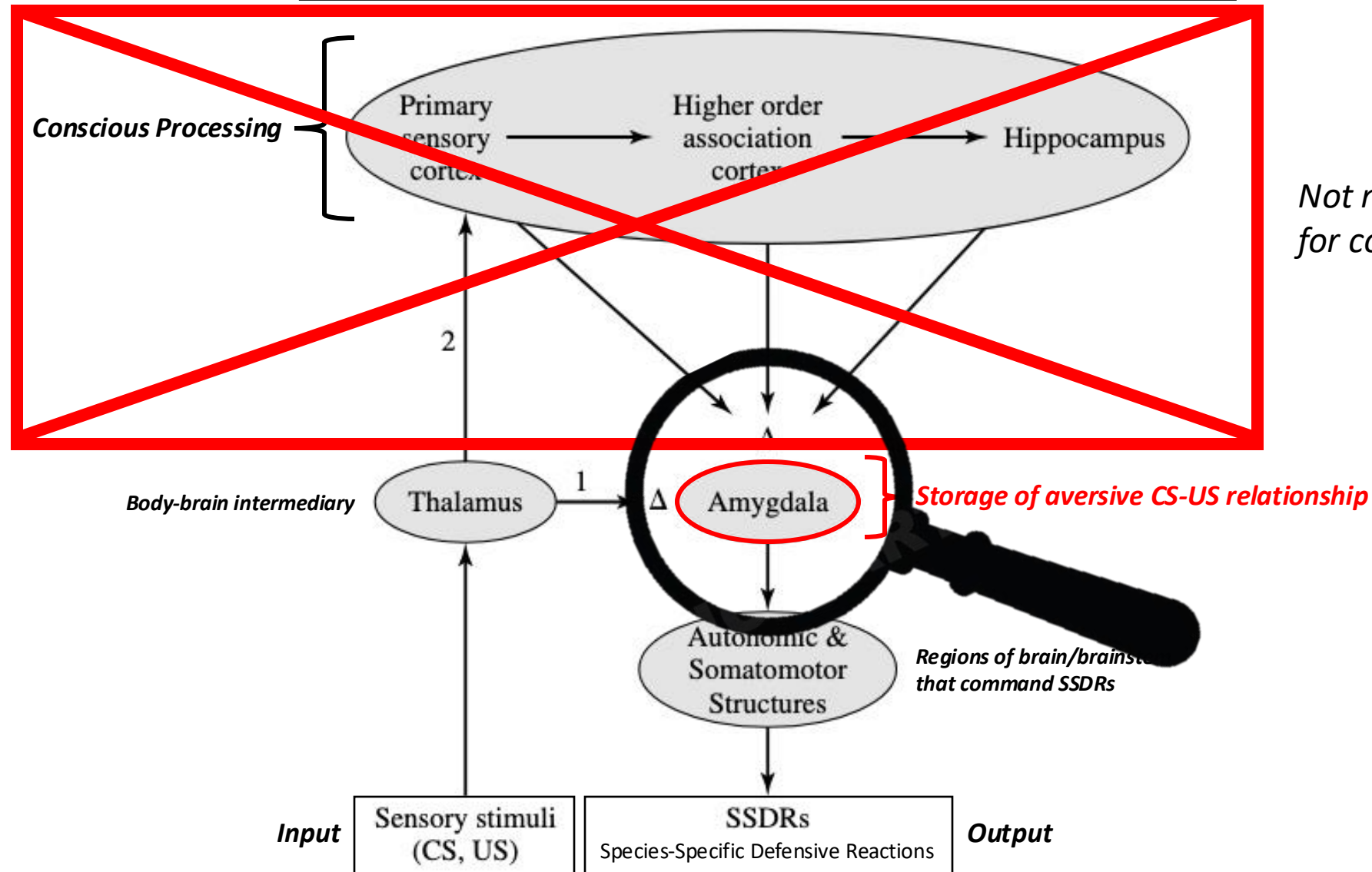
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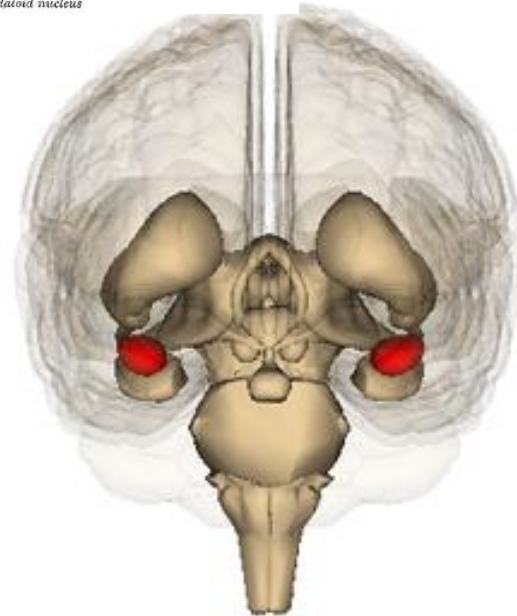
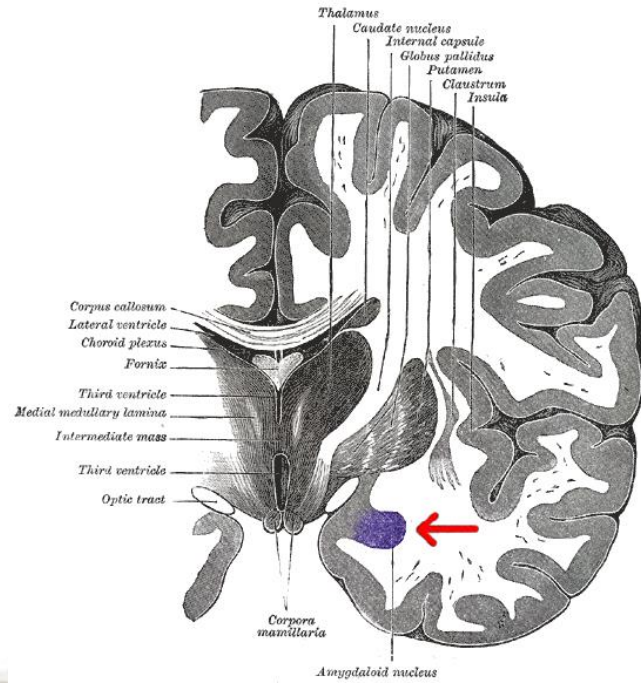


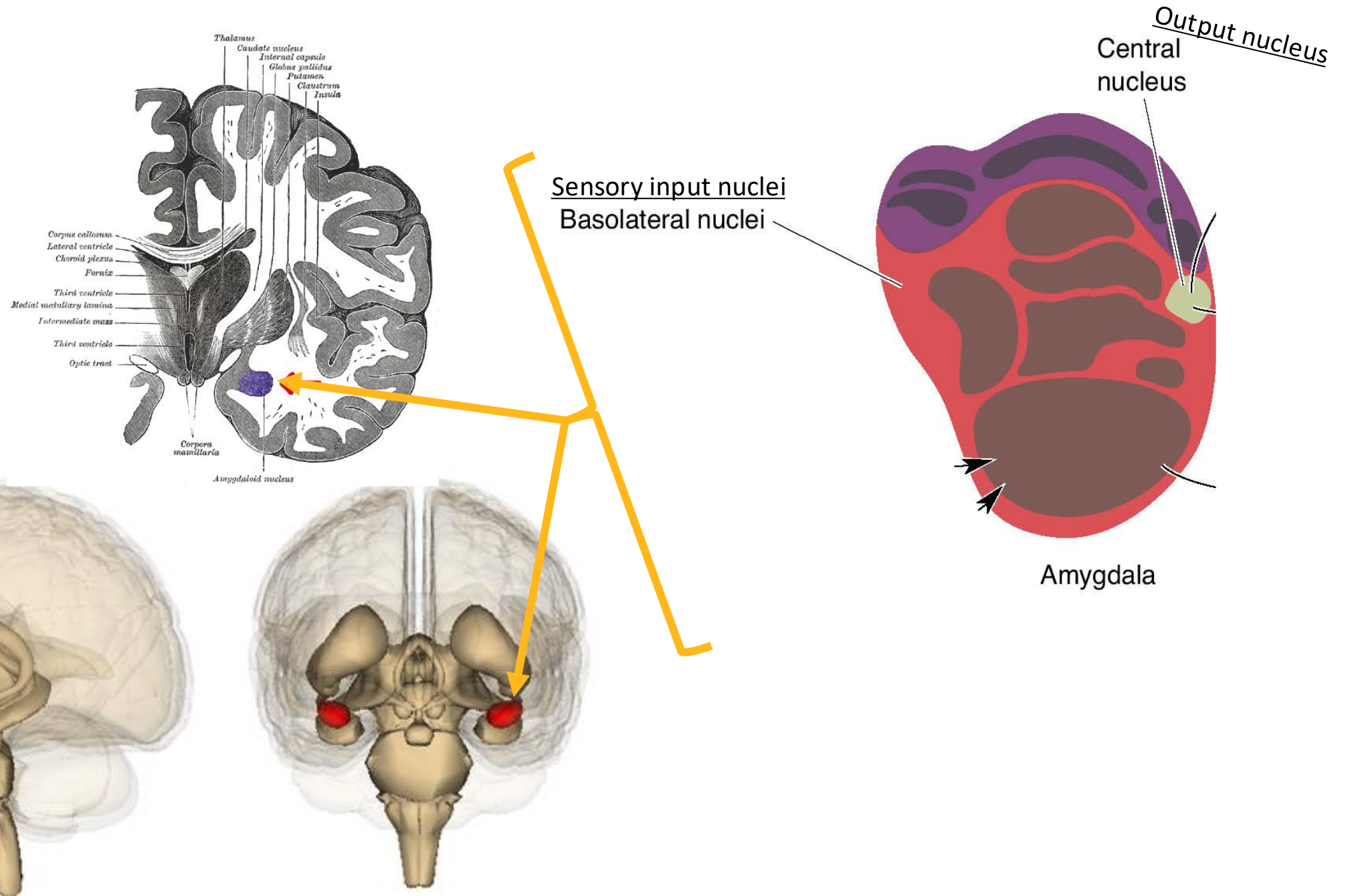
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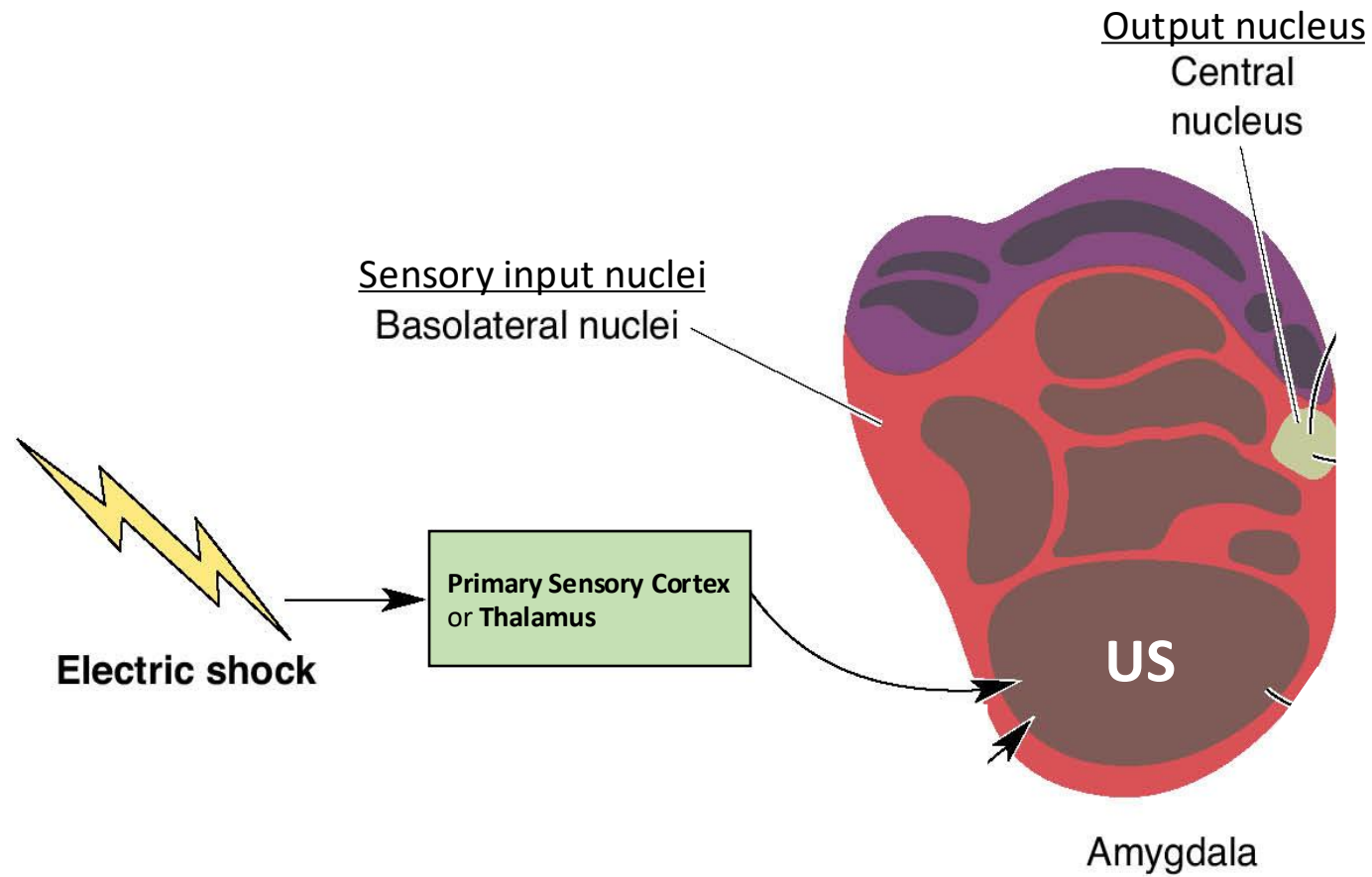


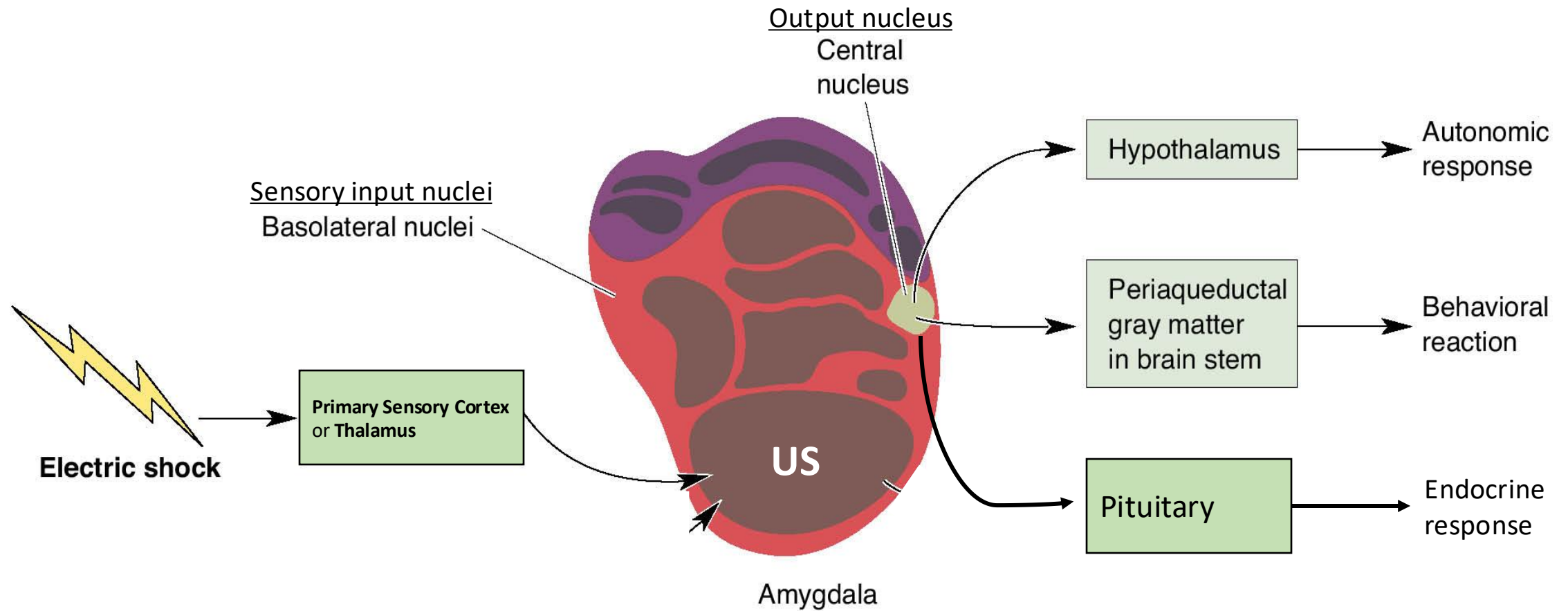
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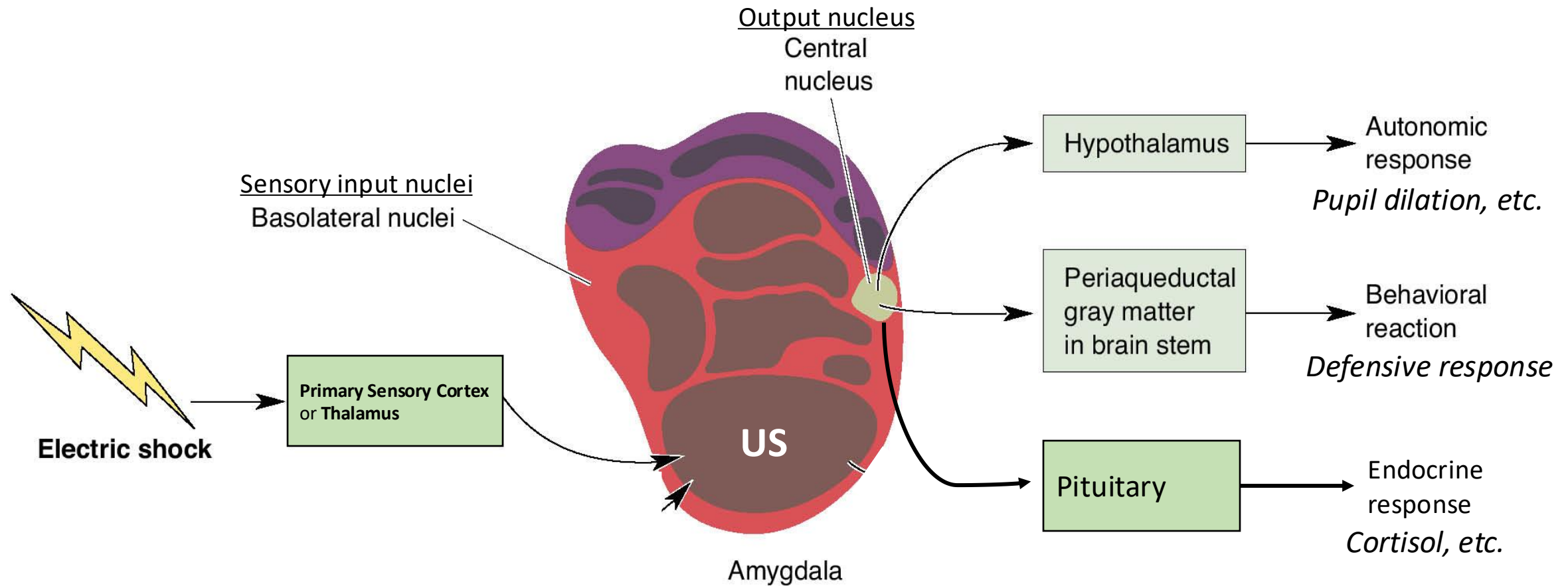


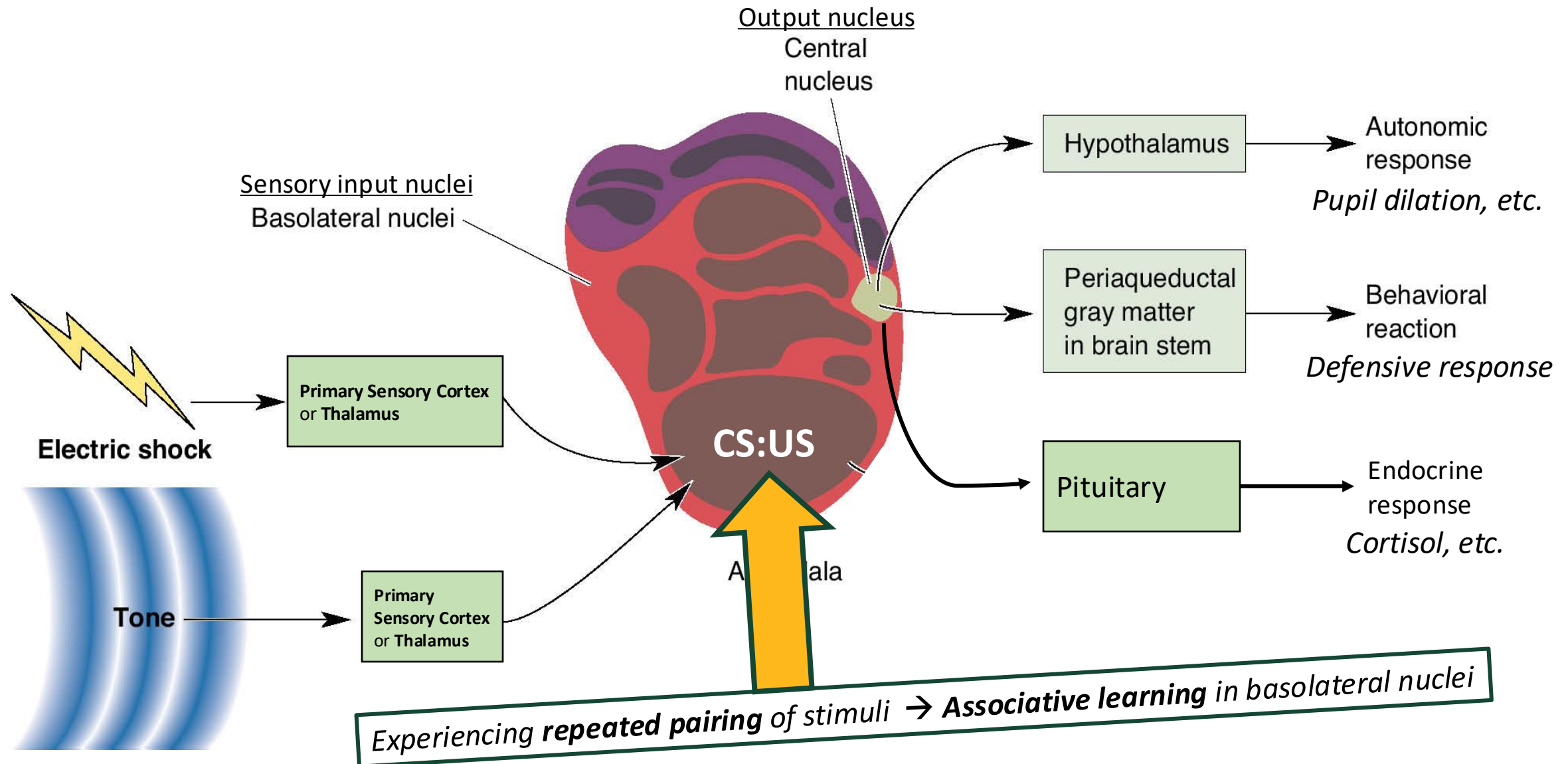


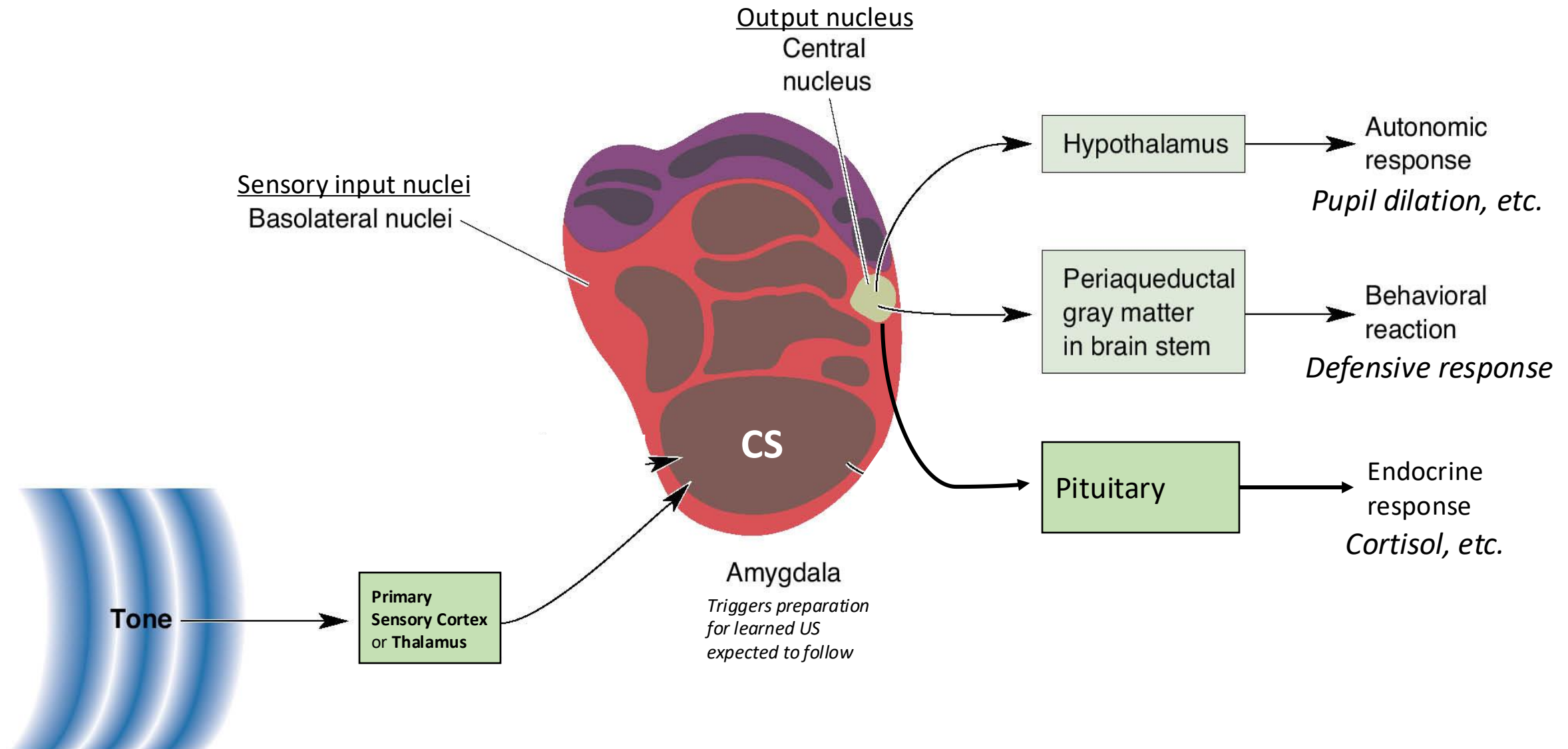


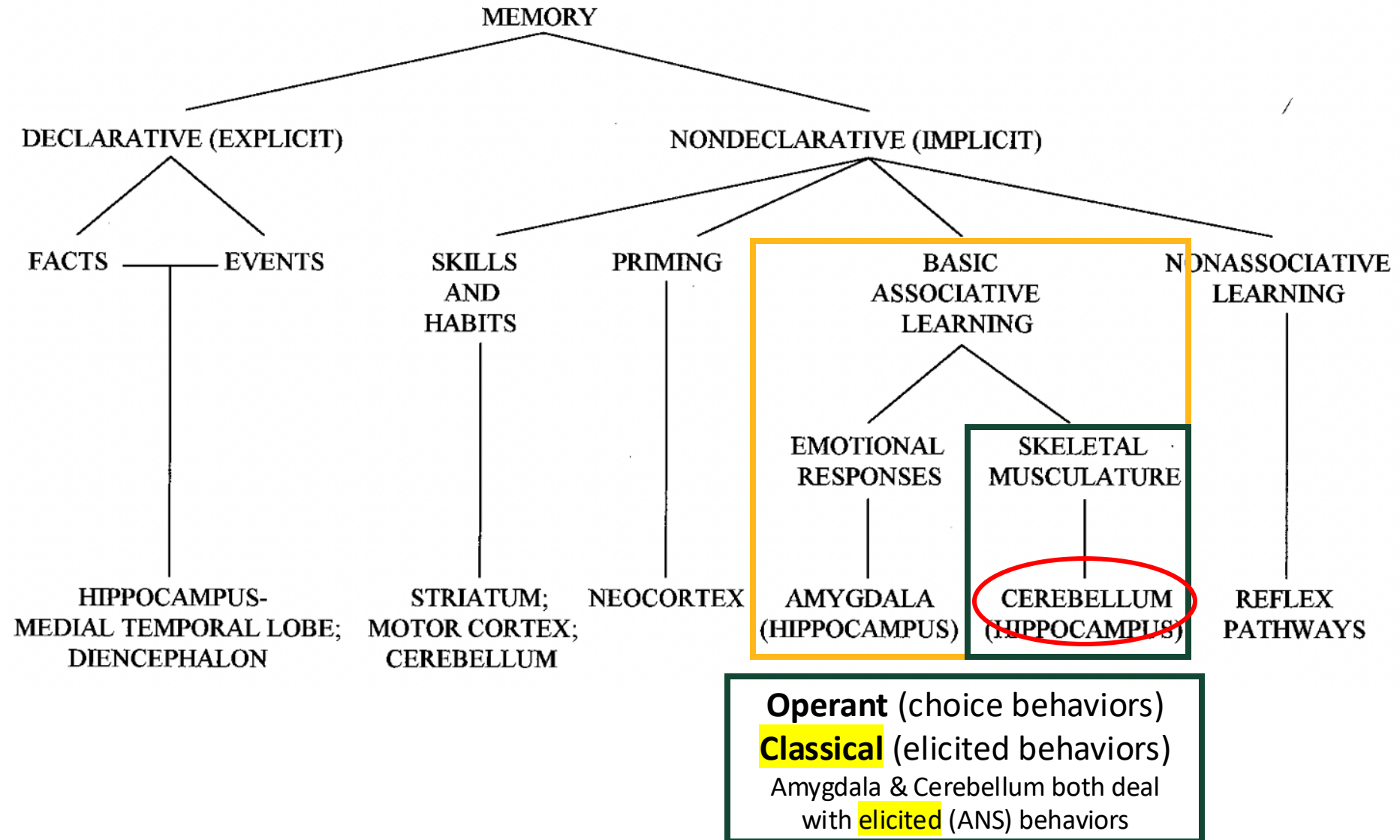


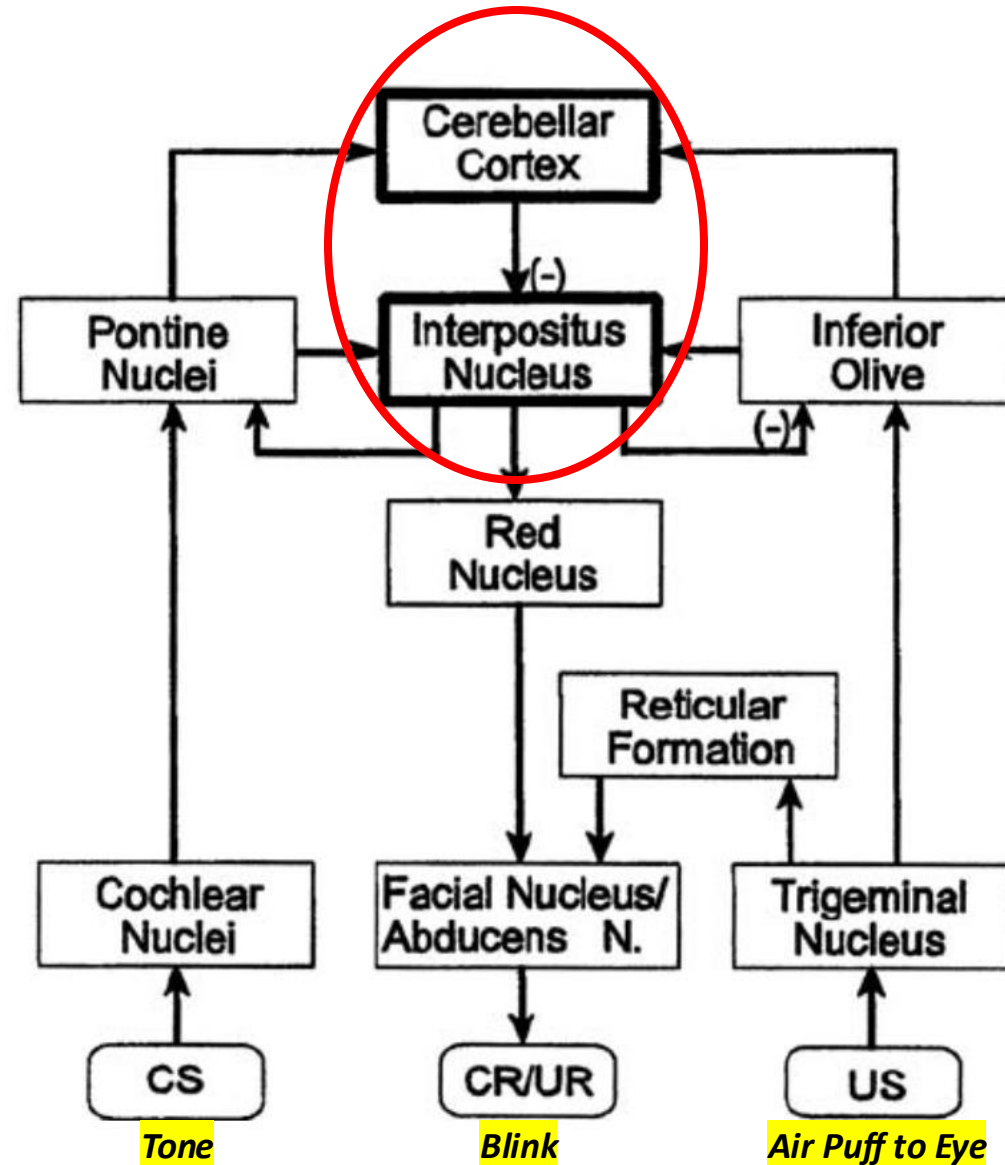






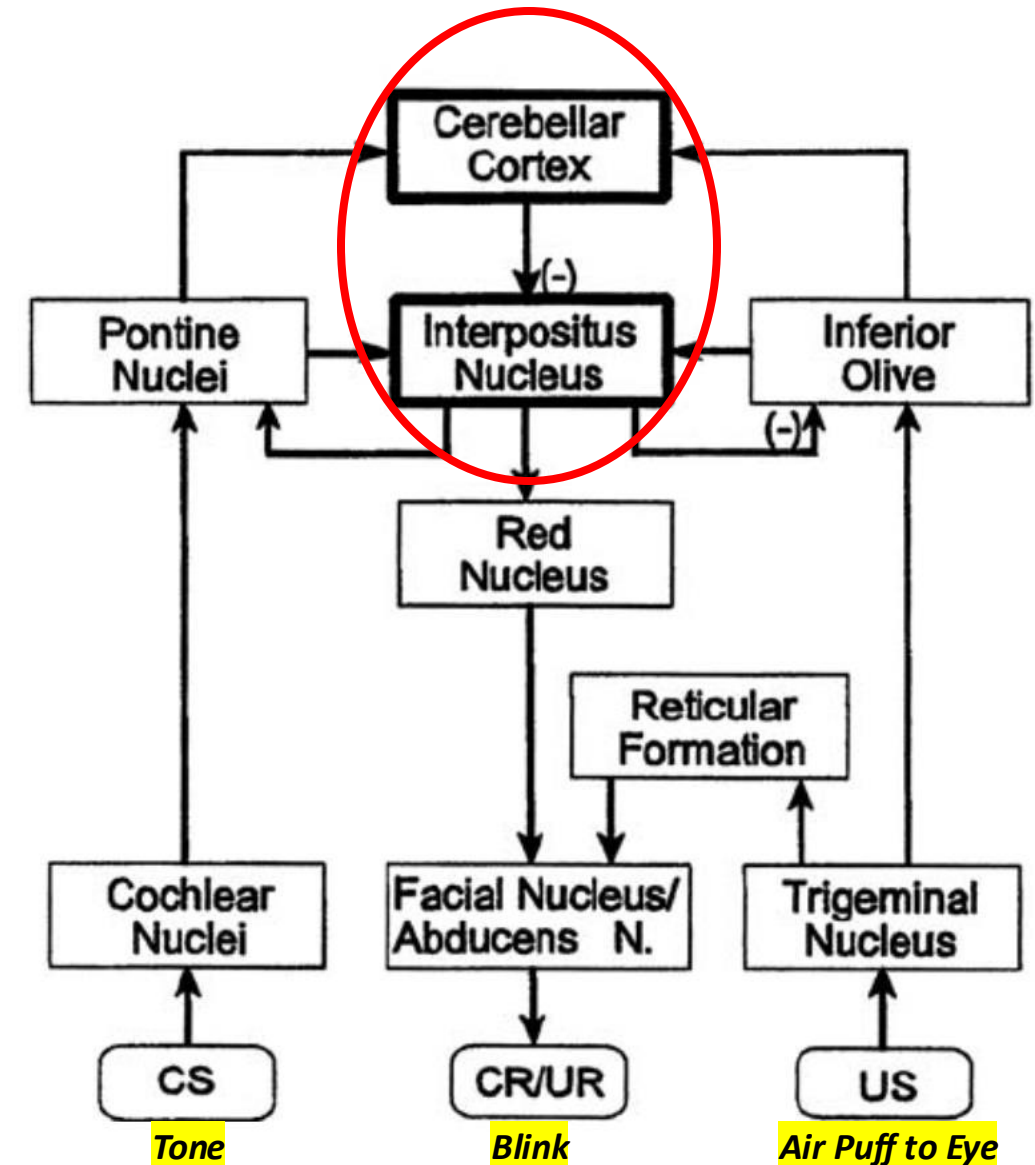


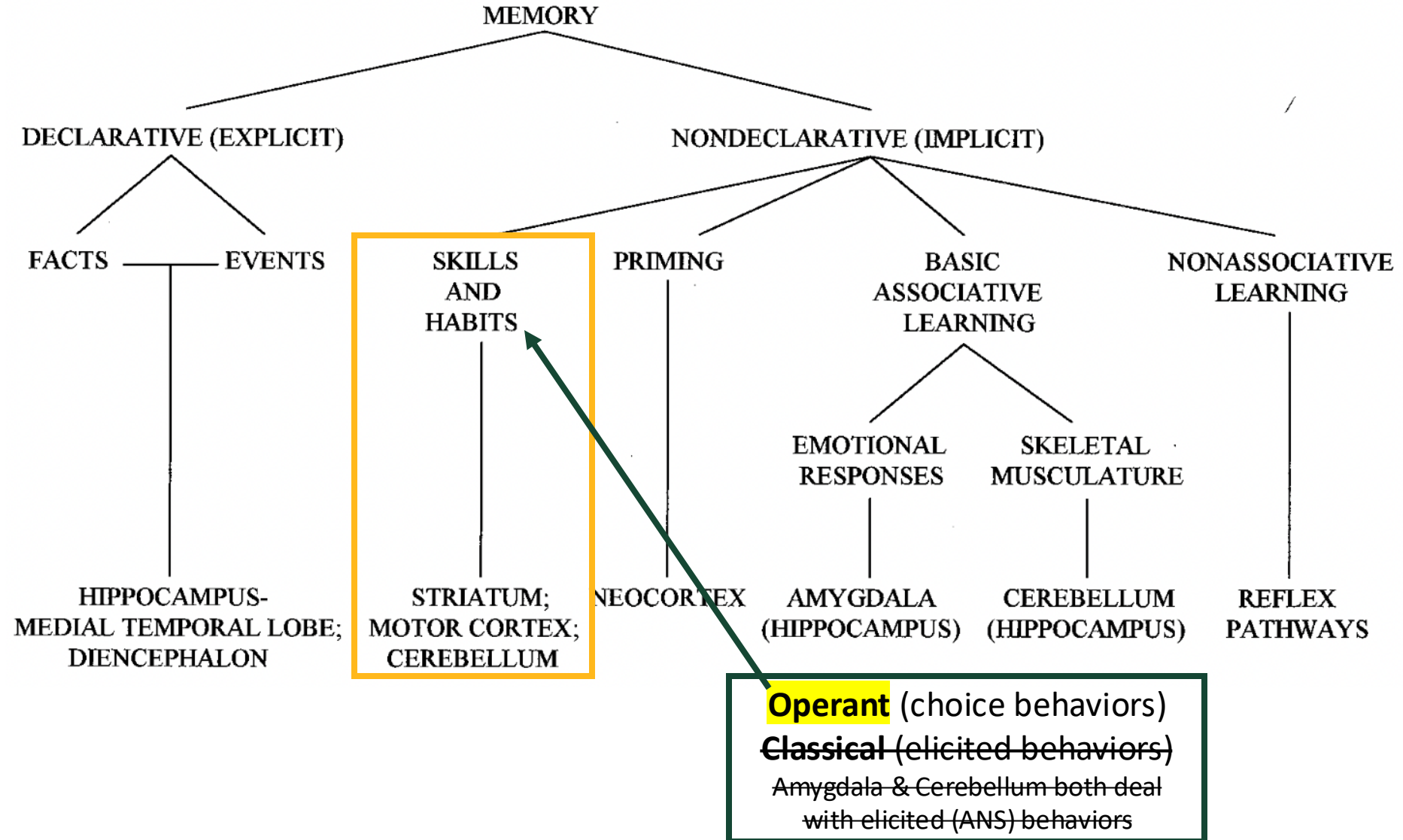




Exact functions of Cerebellum in this circuit an area of ongoing research, but it mediates the CS:US learning in a conditioning paradigm where the response is a reflexive behavioral response (like a blink or a twitch)

Takeaway:
Cerebellum mediates non-fear-based CS:US learning

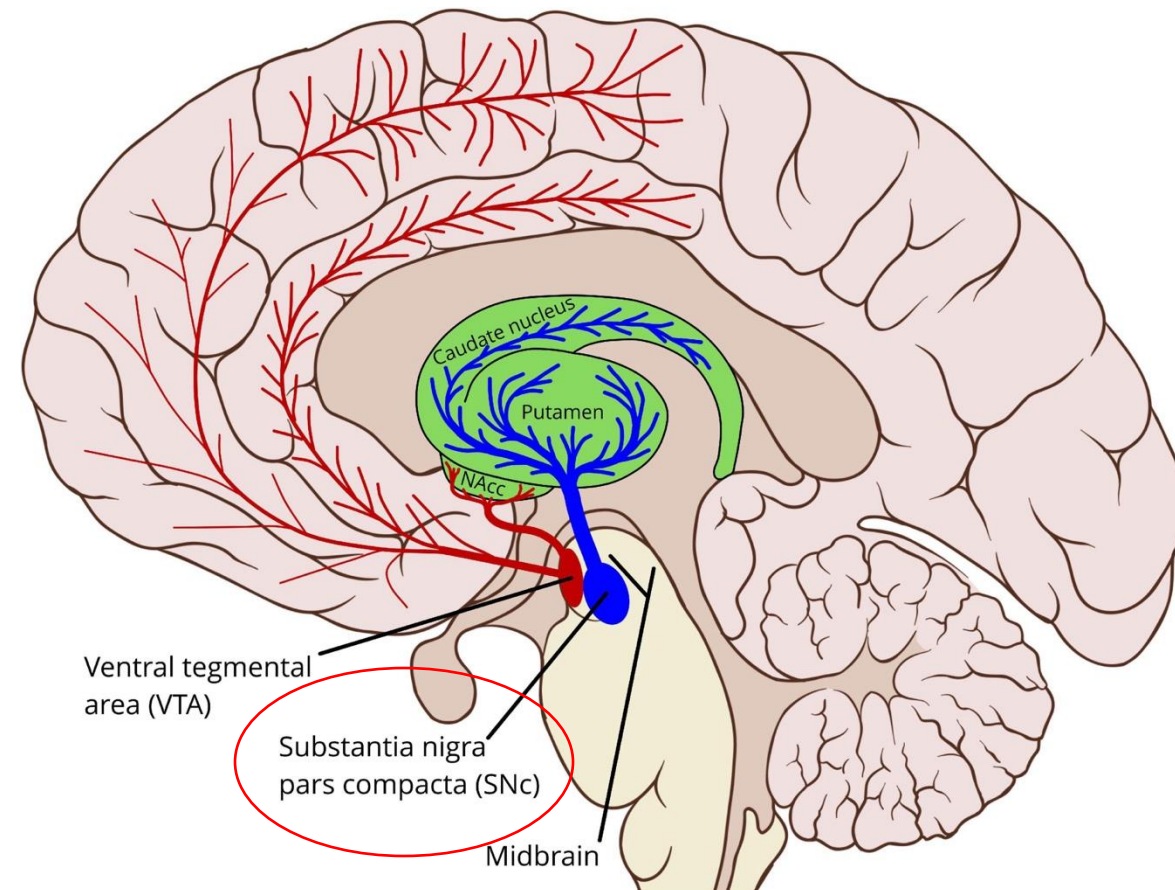




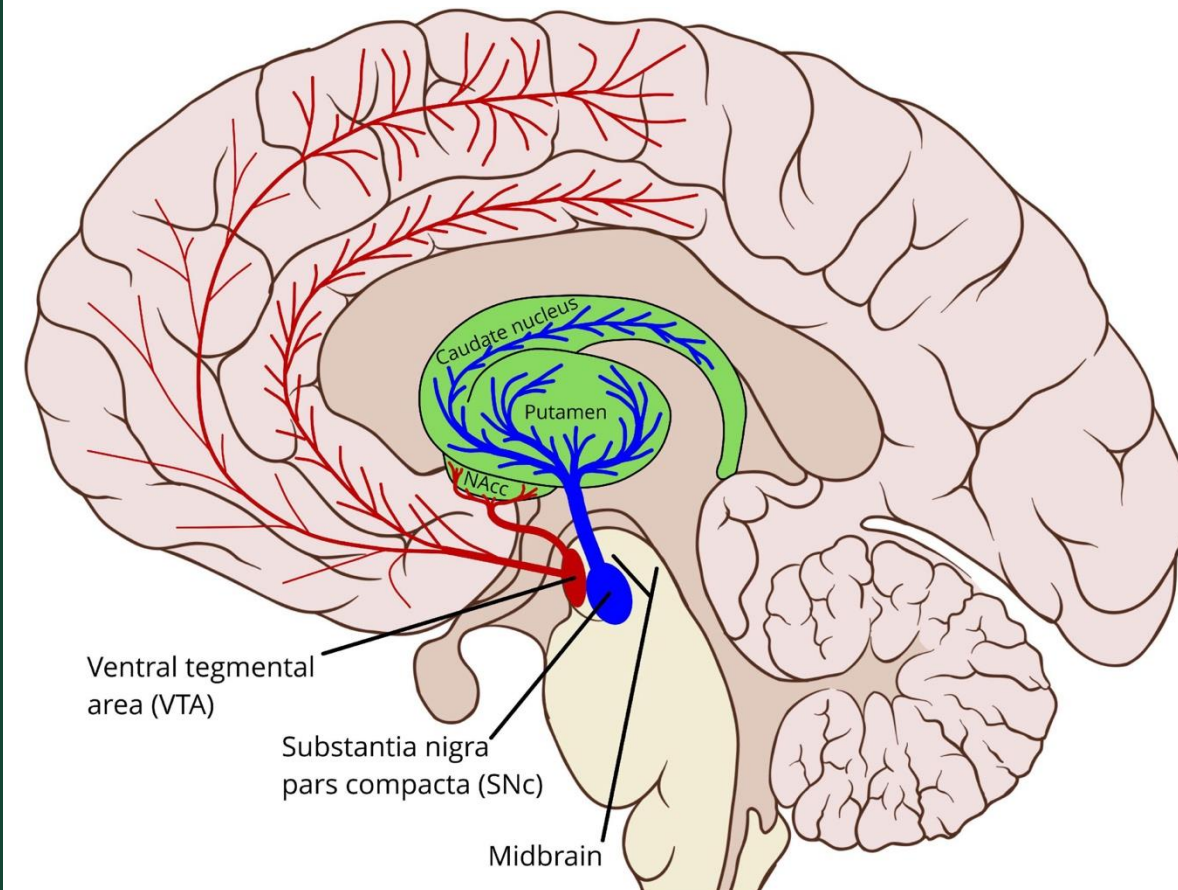
Dopamine and Habit Learning

“The midbrain dopamine system is connected anatomically and functionally with the planning, choosing, initiation, and execution of “voluntary” movements” (Maiorov, 2018).

- Dopamine neurons all begin in one of two small areas in the Midbrain
- Dopamine neuron axons extend throughout Frontal lobe and Striatum

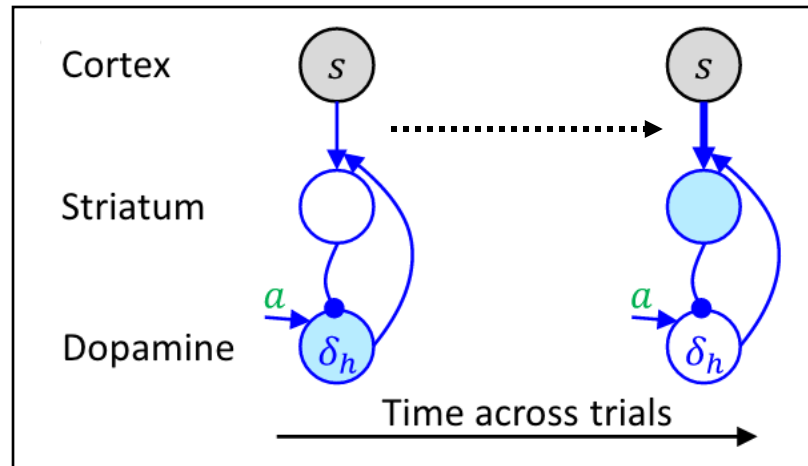


- Dopamine neurons all begin in one of two small areas in the Midbrain
- Dopamine neuron axons extend throughout Frontal lobe and Striatum
- When dopamine neurons fire, they flood these areas with dopamine
- Dopamine neurons fire to potentiate activity in an area



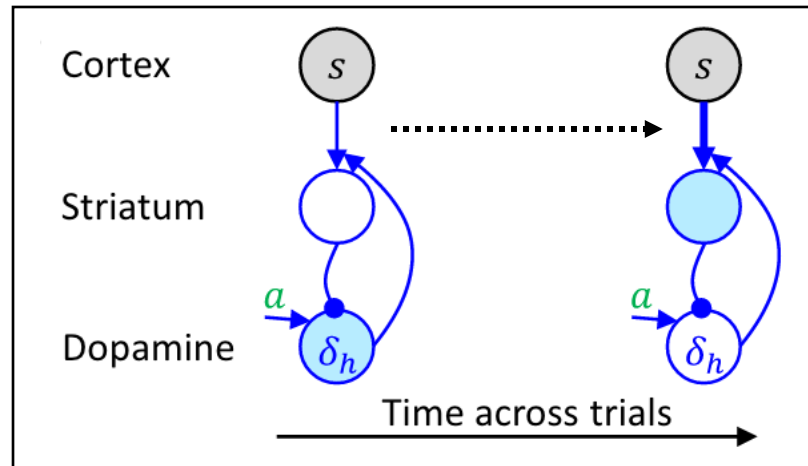
A simple model of the key neurobiology in habit-building

Operant Conditioning can lead to habits, so this is also the biology of Operant Conditioning



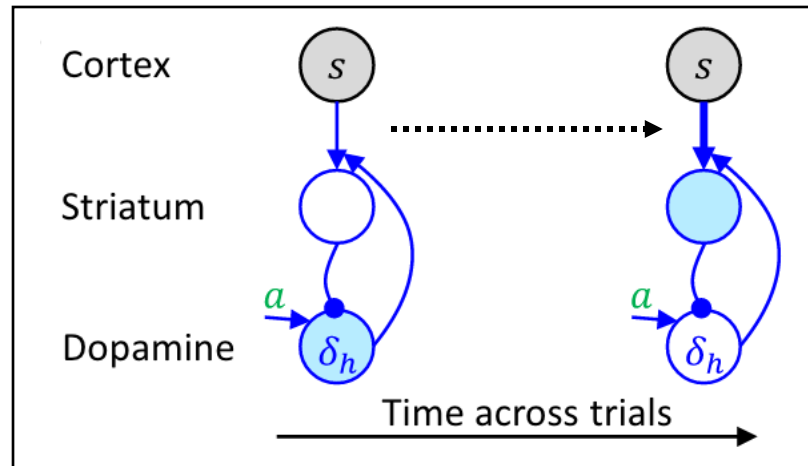
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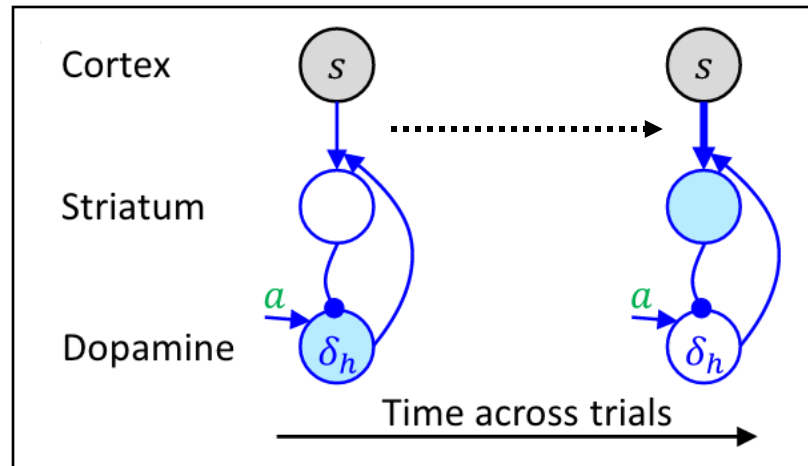
Habit:

actions without considering the value of the current reward available



“Habit System”

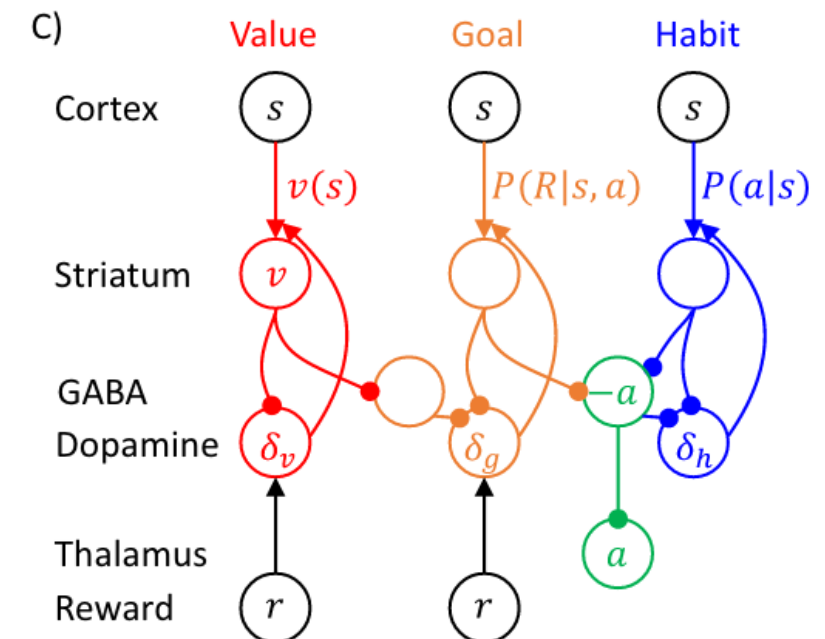
Dopamine triggers the Striatum in Corticostriatal circuits to consolidate the information that in ‘ s ’ state, ‘ a ’ action should occur [because ‘ a ’ action achieves the goal of attaining some outcome of positive value (a reward)]

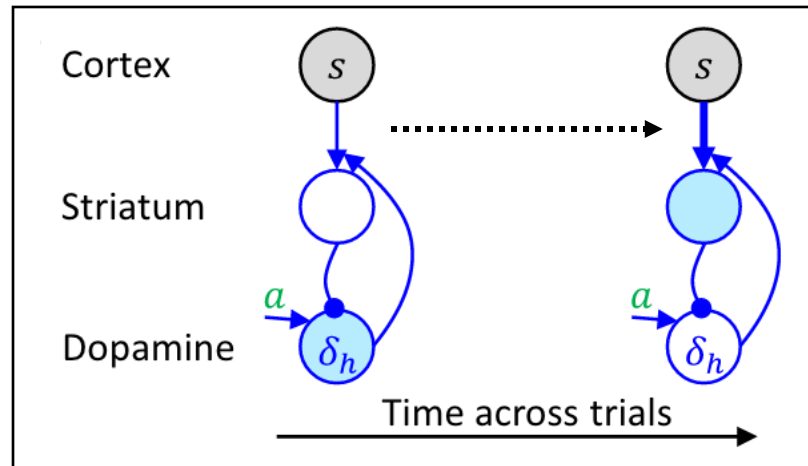


“Habit System”

Dopamine triggers the Striatum in Corticostriatal circuits to consolidate the information that in ‘*s*’ state, ‘*a*’ action should occur [because ‘*a*’ action achieves the goal of attaining some outcome of positive value (a reward)]

Does the **Goal system’s action plan** (which is based on what Action it predicts will get the Reward in this State) **match the Habit system’s action plan** (which is based on what Action is associated with [i.e., typically occurs in response to] this State)?



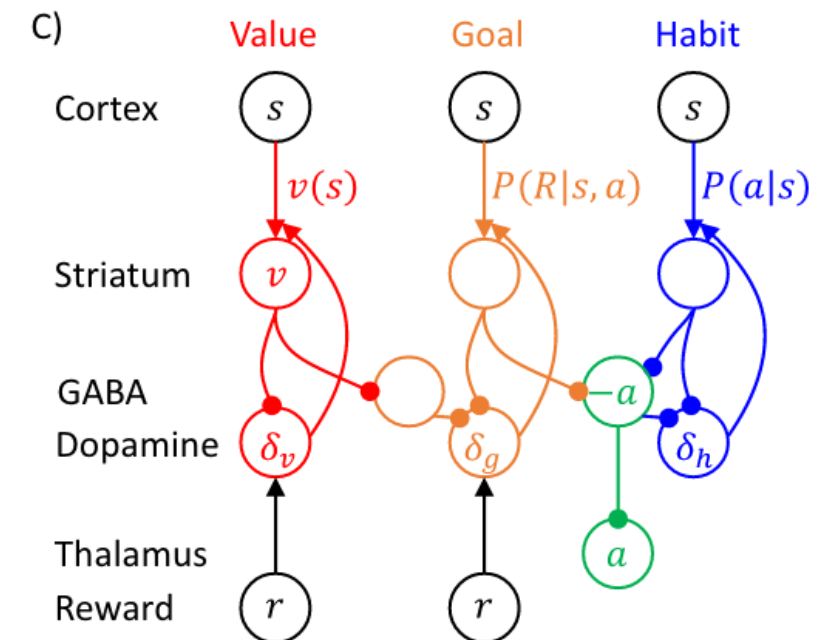


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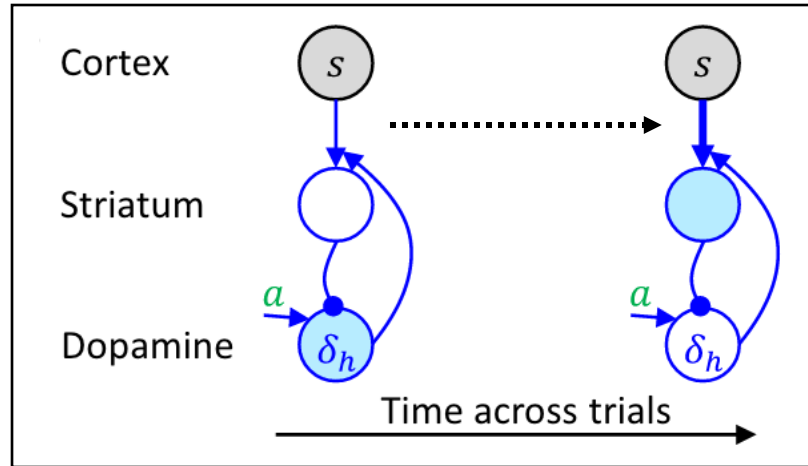
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“habits are formed through a process in which the **habit system learns to mimic the goal-directed system**”



A simple model of the key neurobiology in habit-building

Operant Conditioning can lead to habits, so this is also the biology of Operant Conditioning



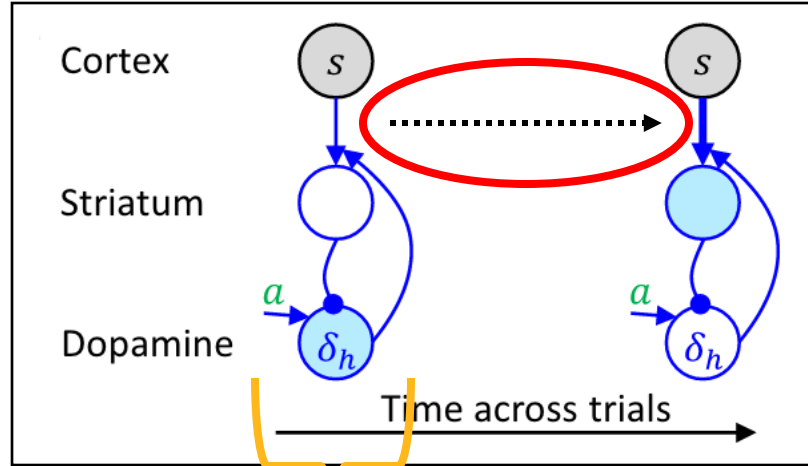
"Habit System"

Dopamine triggers the Striatum in Corticostriatal circuits to consolidate the information that in ' s ' state, ' a ' action should occur [because ' a ' action achieves the goal of attaining some outcome of positive value (a reward)]

Goal system action plan = Habit system action plan?

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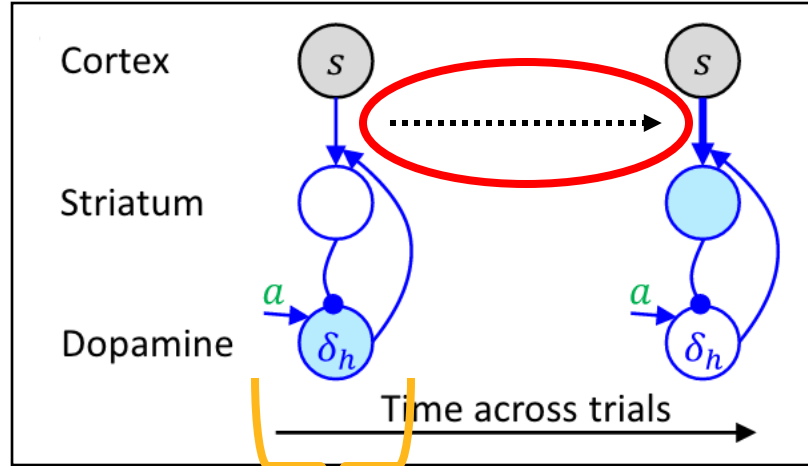
No:

DA system activates. Triggers Habit system to encode (store) that State-Action relationship into the Striatum at Corticostriatal synapses.

Goal system action plan = Habit system action plan?

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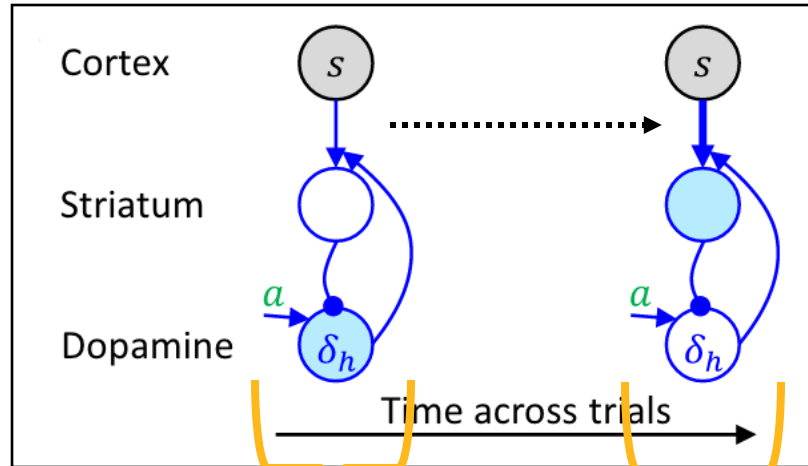
No:

Now, in the future, that State will trigger the Striatum to automatically plan 'a' action.

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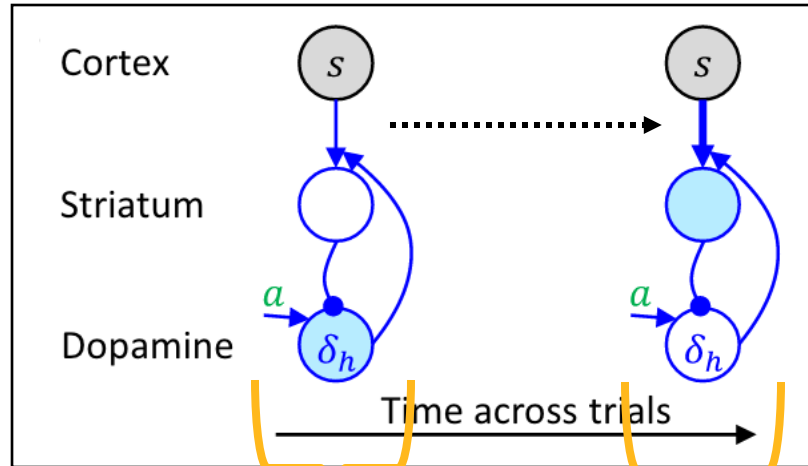
Yes:

State representation in cortex activates Striatum, which plans Action associated with that State. Striatum activity inhibits DA system.

Goal system action plan = Habit system action plan?

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Dopamine triggers the Striatum in Corticostriatal circuits to consolidate the information that in ‘s’ state, ‘a’ action should occur [because ‘a’ action achieves the goal of attaining some outcome of positive value (a reward)]

“At the start of training, when an action is selected mostly by the goal-directed system, the dopaminergic neurons in the habit system receive an input encoding the chosen action, but the striatal neurons in the habit system are not yet able to predict this action, resulting in a prediction error encoded in dopaminergic activity (left display in Figure 1D). This prediction error triggers plasticity in the striatal neurons of the habit system, so they tend to predict this action in the future (right display in Figure 1D).” (Bogacz, 2020).

No:

DA system activates. Triggers Habit system to encode (store) that State-Action relationship into the Striatum at Corticostriatal synapses.

Now, in the future, that State will trigger the Striatum to automatically plan ‘a’ action.

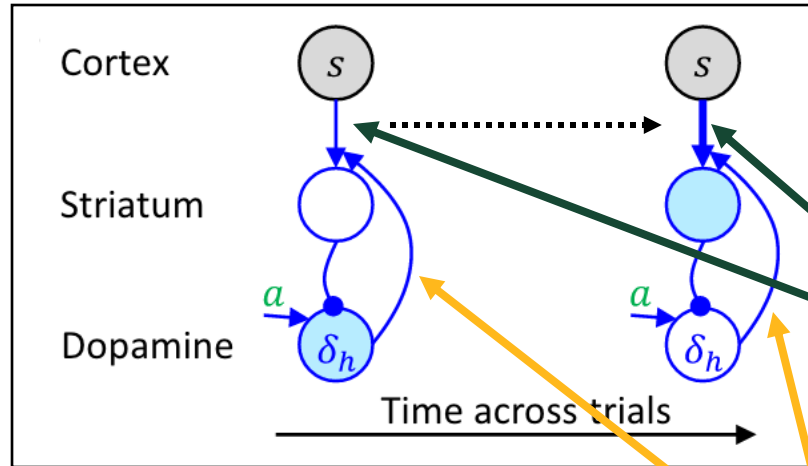
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Goal system action plan = Habit system action plan?

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“Habit System”

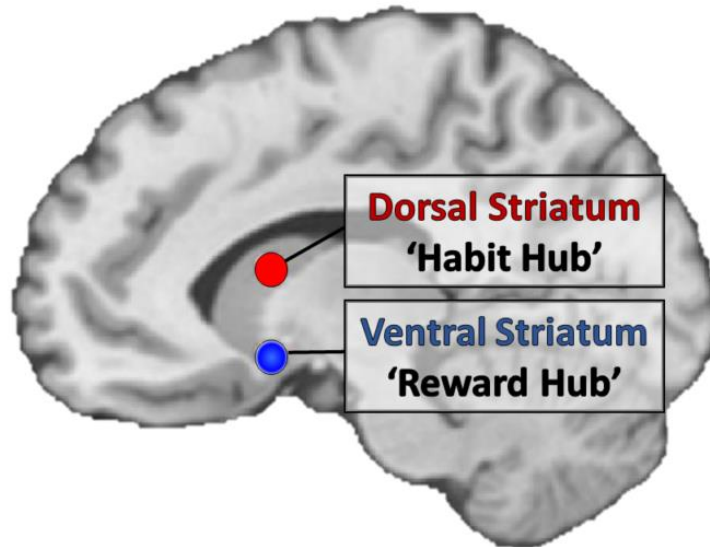
Dopamine triggers the **dorsal** Striatum in Corticostriatal circuits to consolidate the information that in ‘s’ state, ‘a’ action should occur [because ‘a’ action achieves the goal of attaining some outcome of positive value (a reward)]

Corticostriatal Pathway

Frontal Lobe $\rightarrow$ Dorsal Striatum

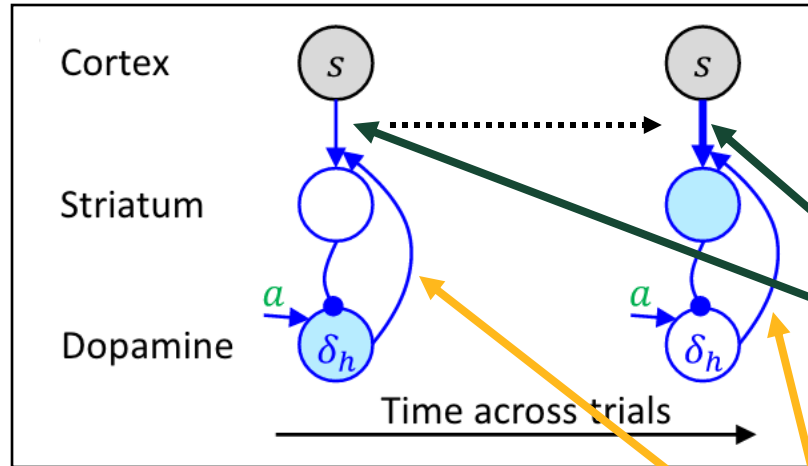
Nigrostriatal Pathway

Substantia Nigra $\rightarrow$ Dorsal Striatum



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Corticostriatal Pathway

Frontal Lobe $\rightarrow$ Dorsal Striatum

Nigrostriatal Pathway

Substantia Nigra $\rightarrow$ Dorsal Striatum

“Habit System”

Dopamine triggers the dorsal Striatum in Corticostriatal circuits to consolidate the information that in ‘s’ state, ‘a’ action should occur [because ‘a’ action achieves the goal of attaining some outcome of positive value (a reward)]

CORONAL SECTION - MYELIN STAIN
(DARK AREAS ARE WHITE MATTER)

CAUDATE

2-MINUTE
NEUROSCIENCE

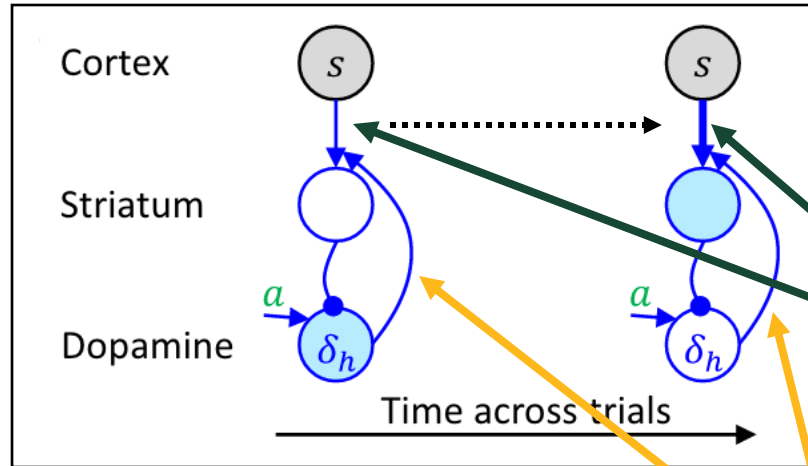
PUT

STRIATUM

NUCLEUS ACCUMBENS

A simple model of the key neurobiology in habit-building

Operant Conditioning can lead to habits, so this is also the biology of Operant Conditioning

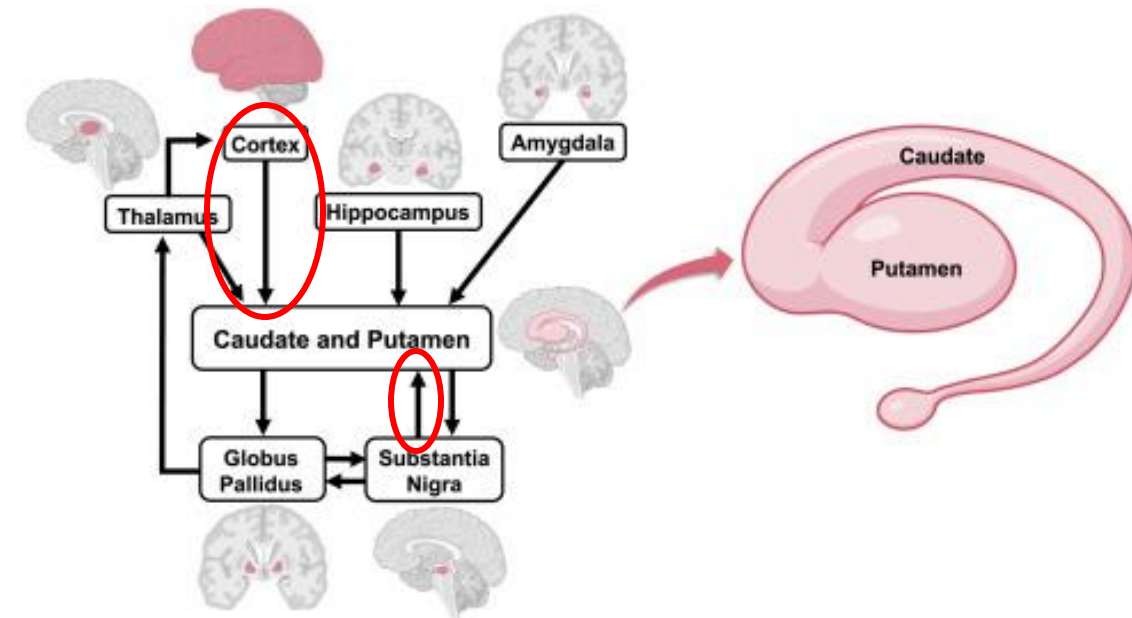
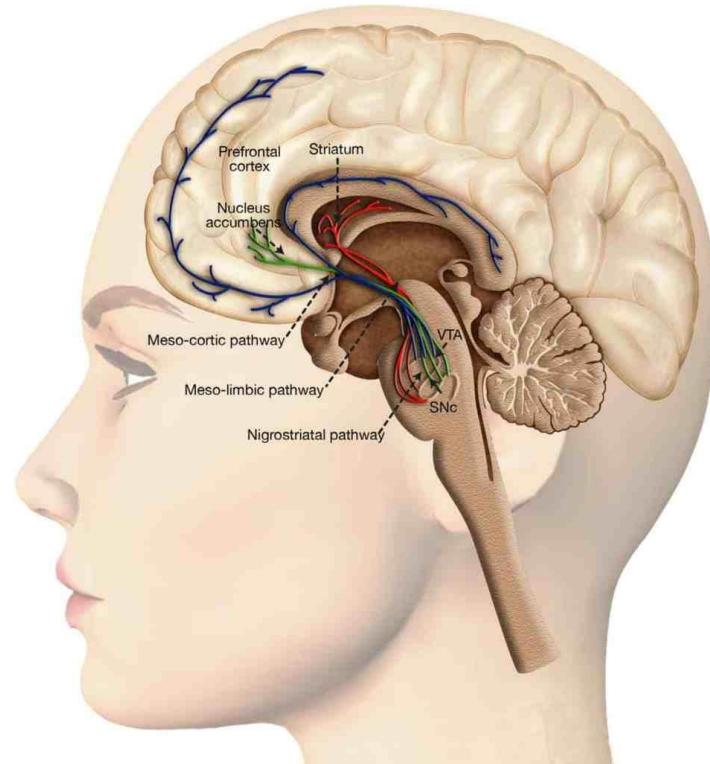


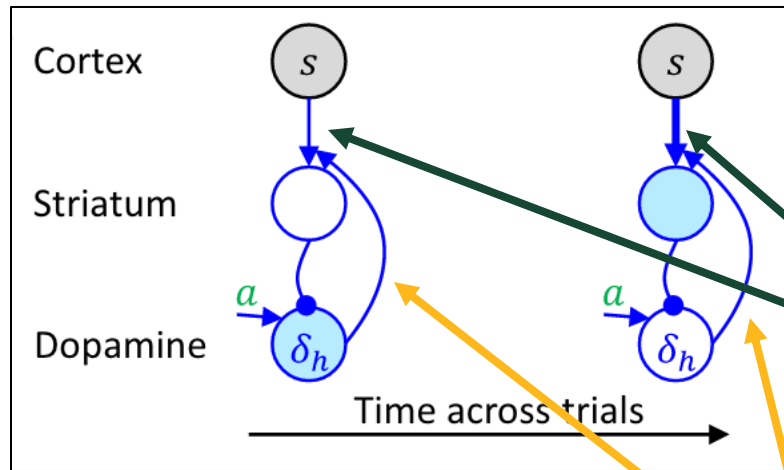
"Habit System"

Dopamine triggers the dorsal Striatum in Corticostriatal circuits to consolidate the information that in 's' state, 'a' action should occur [because 'a' action achieves the goal of attaining some outcome of positive value (a reward)]

Corticostriatal Pathway

Nigrostriatal Pathway

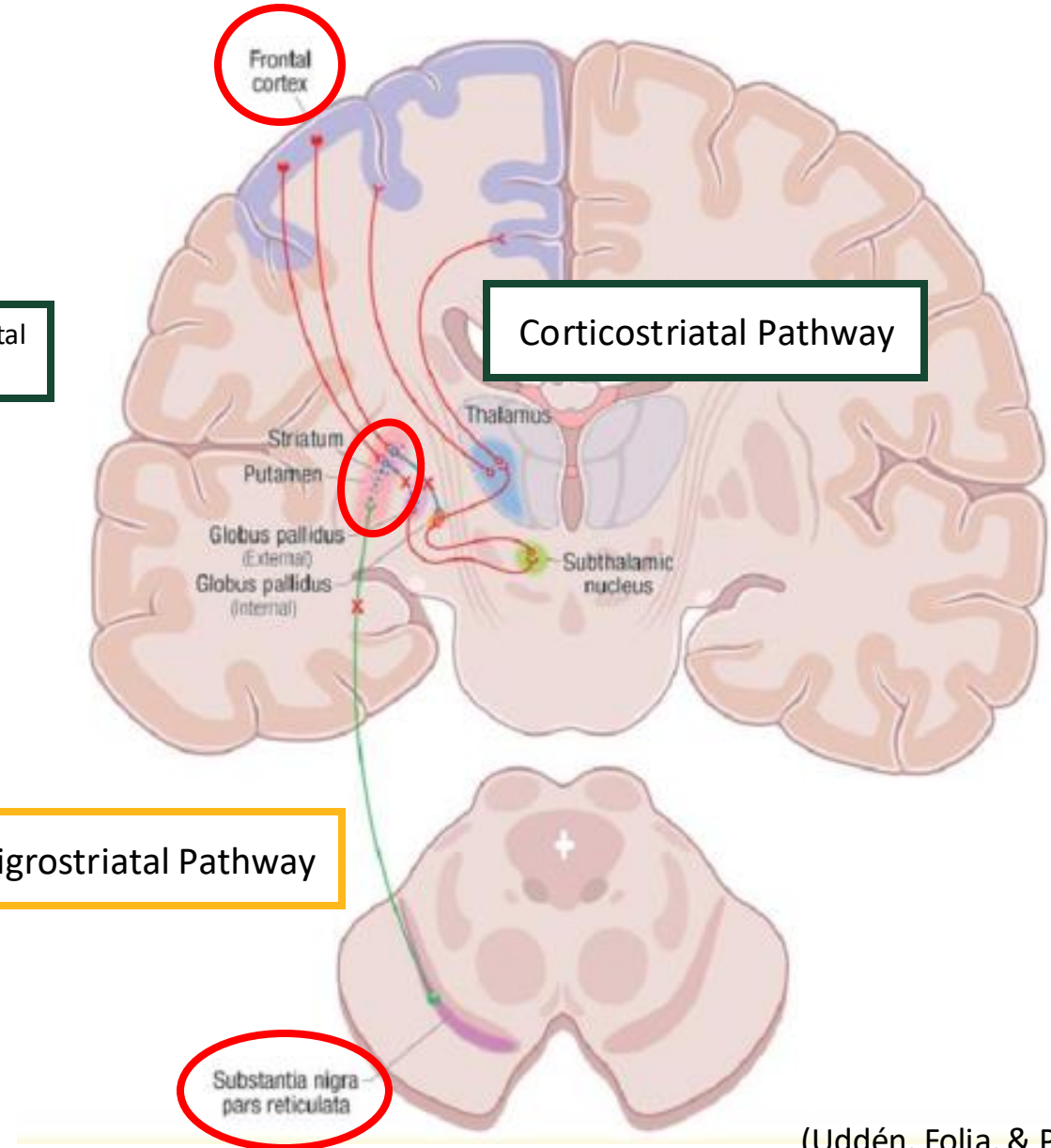


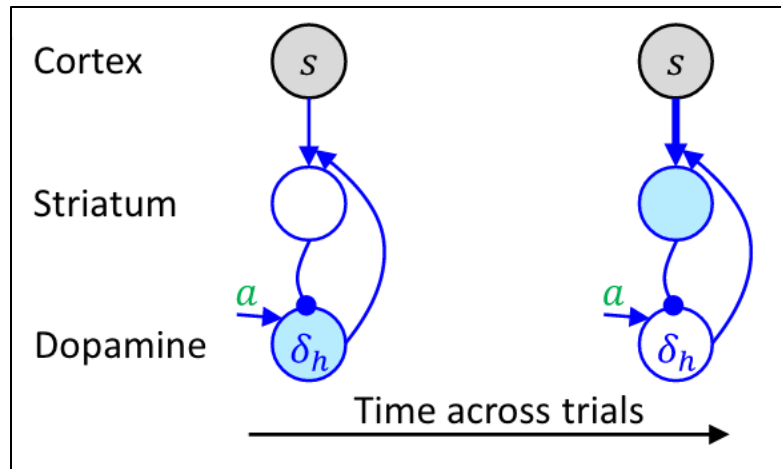


Corticostriatal Pathway

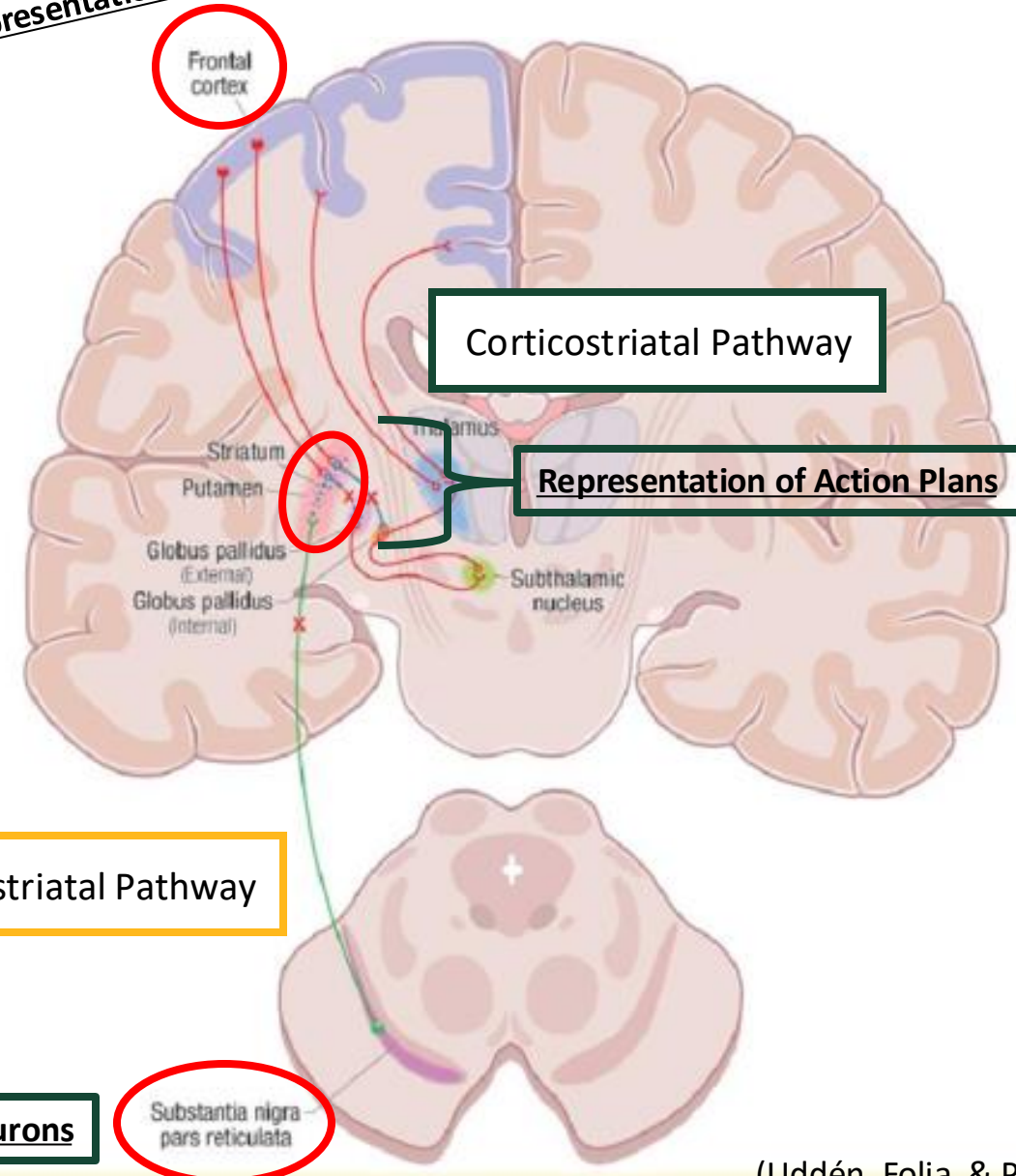
Nigrostriatal Pathway

Nigrostriatal Pathway





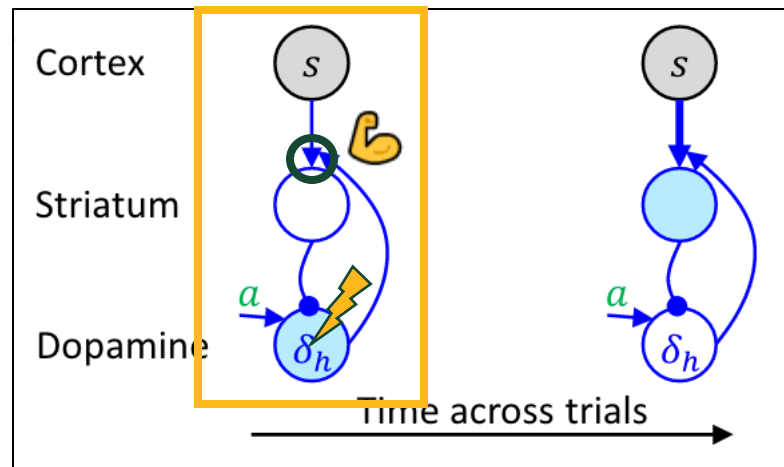
Representation of Environment ('S')



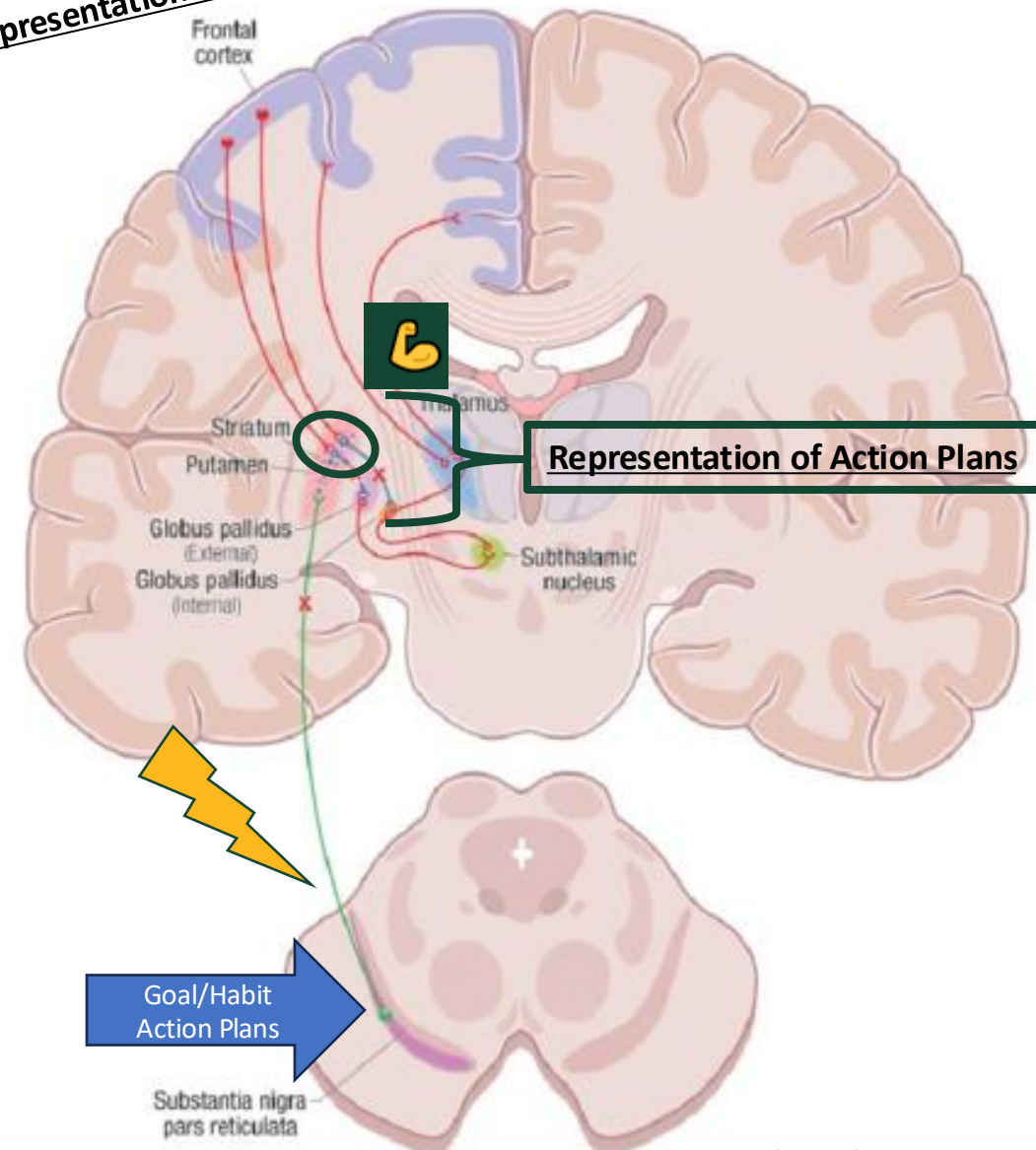
Origin of Dopamine neurons

Substantia nigra
pars reticulata

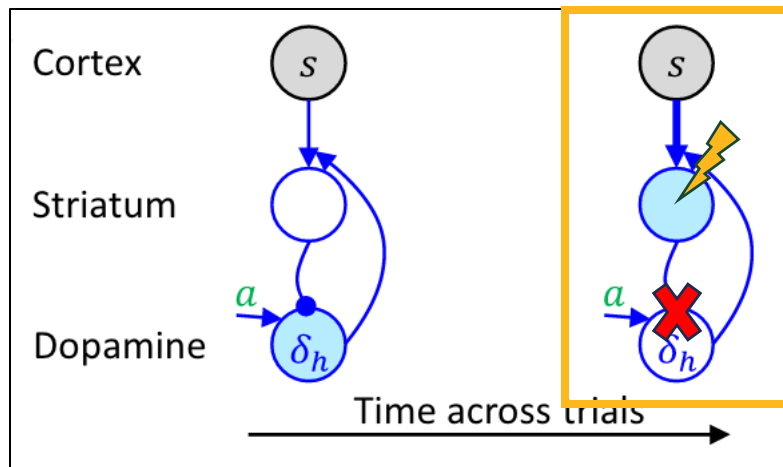
During the first few trials of operant conditioning or the first few instances of learning a habit (*when goal-directed system and habit system action plans don't match*)...



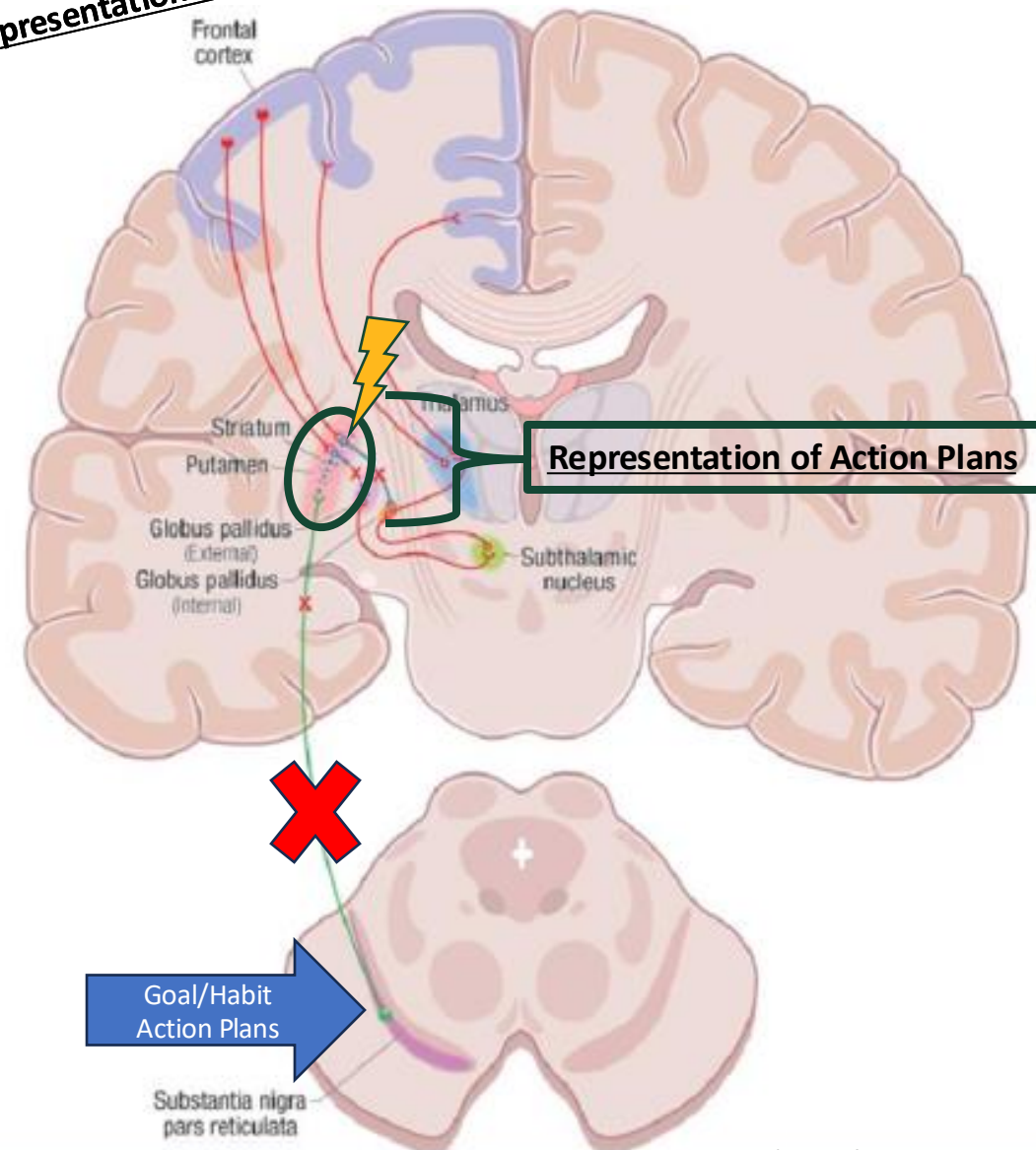
Representation of Environment ('S')

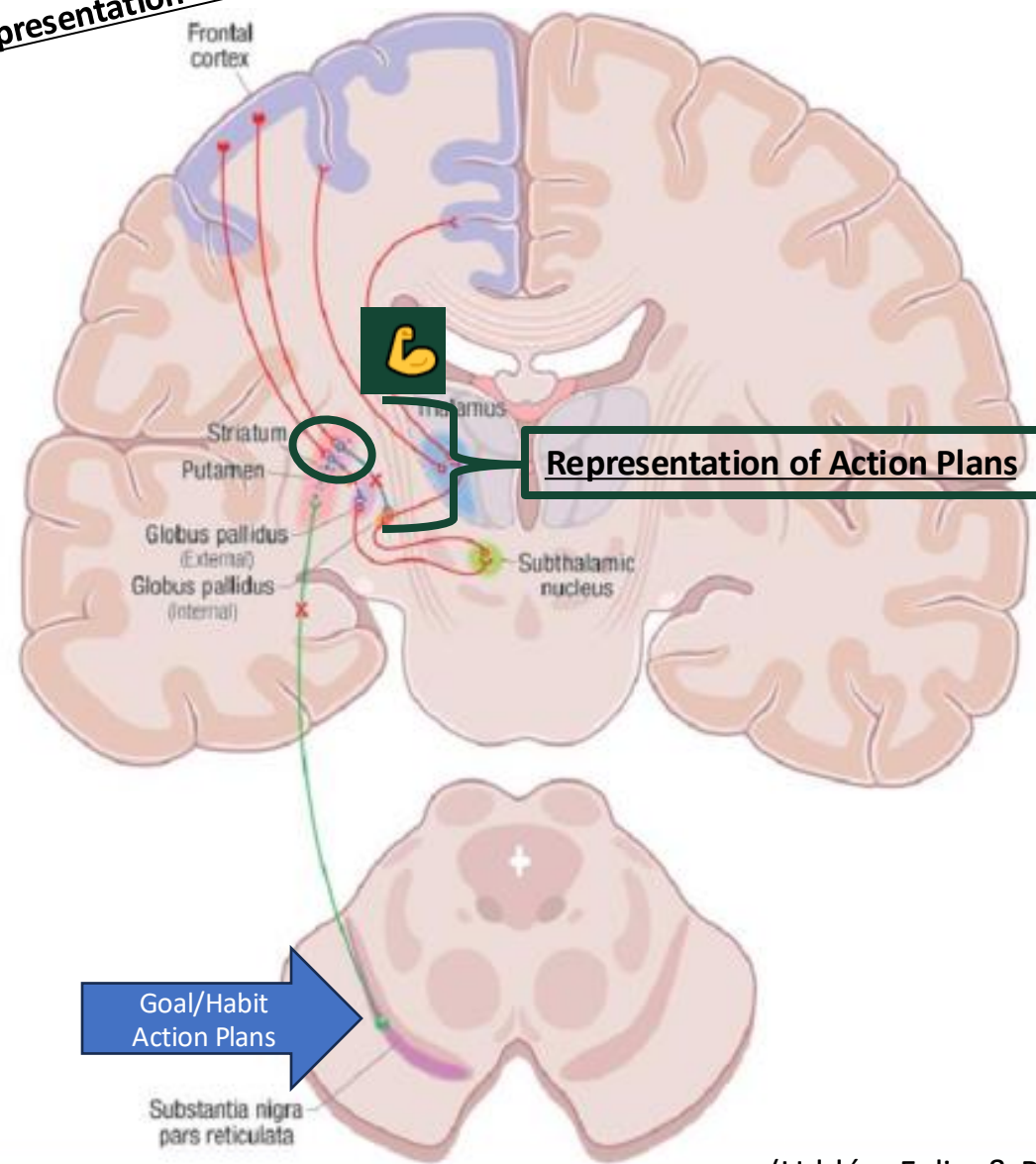
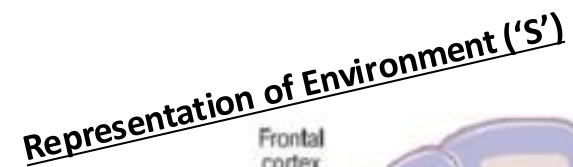
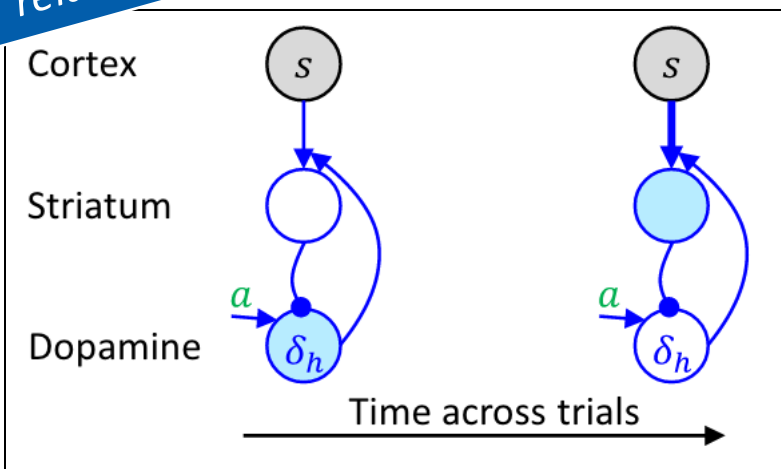
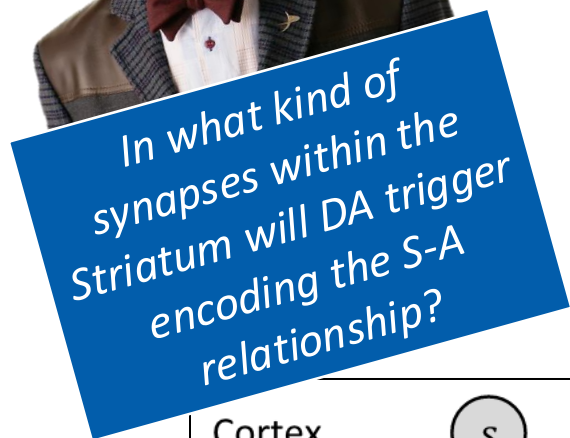


Once the operant conditioning and/or habit is learned (*when habit system mimics goal-directed system*)...



Representation of Environment ('S')

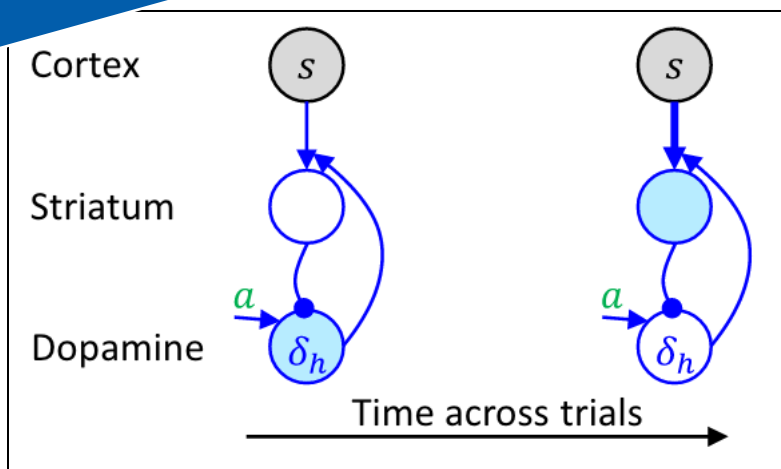




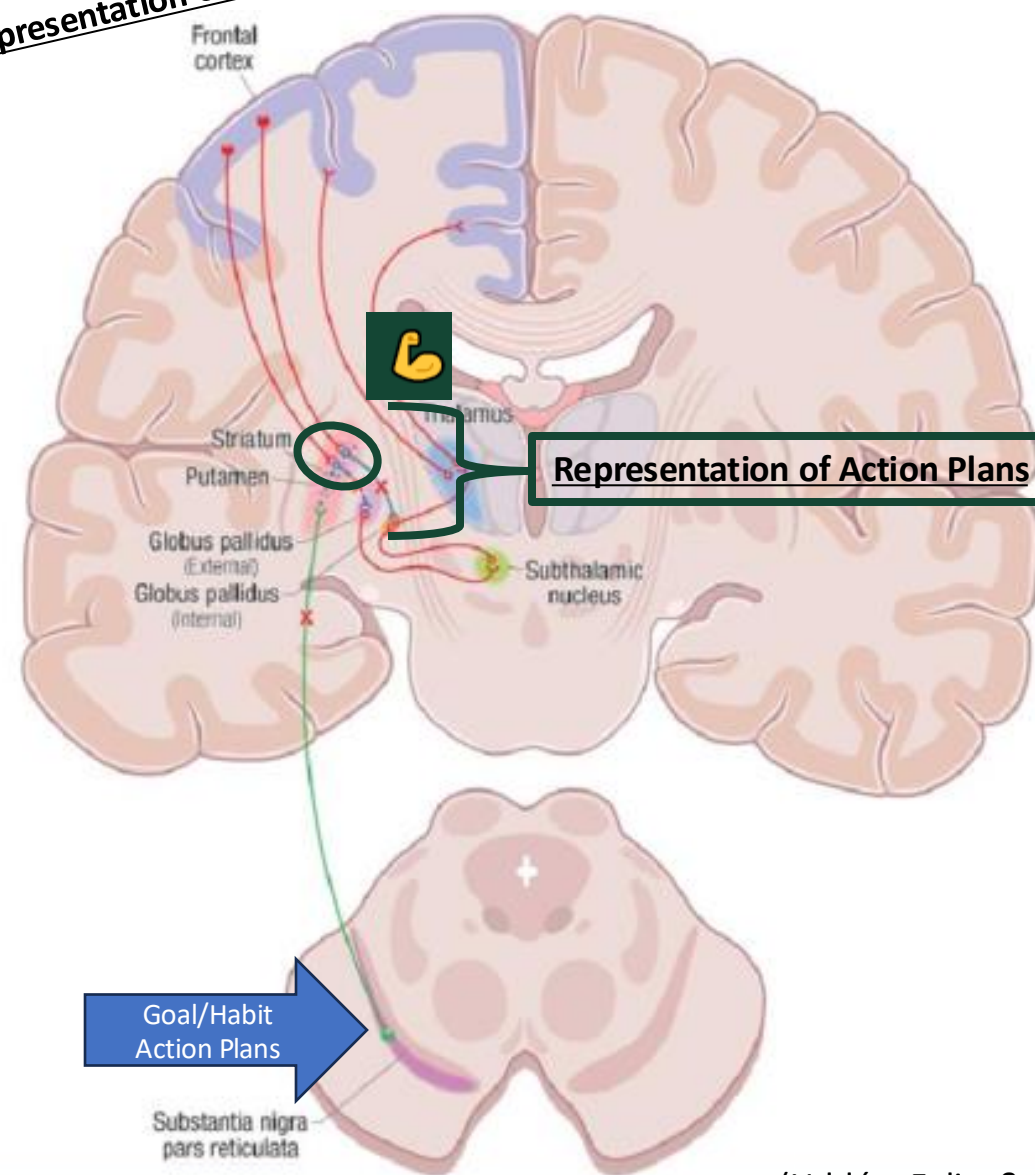
(Uddén, Folia, & Petersson, 2010)

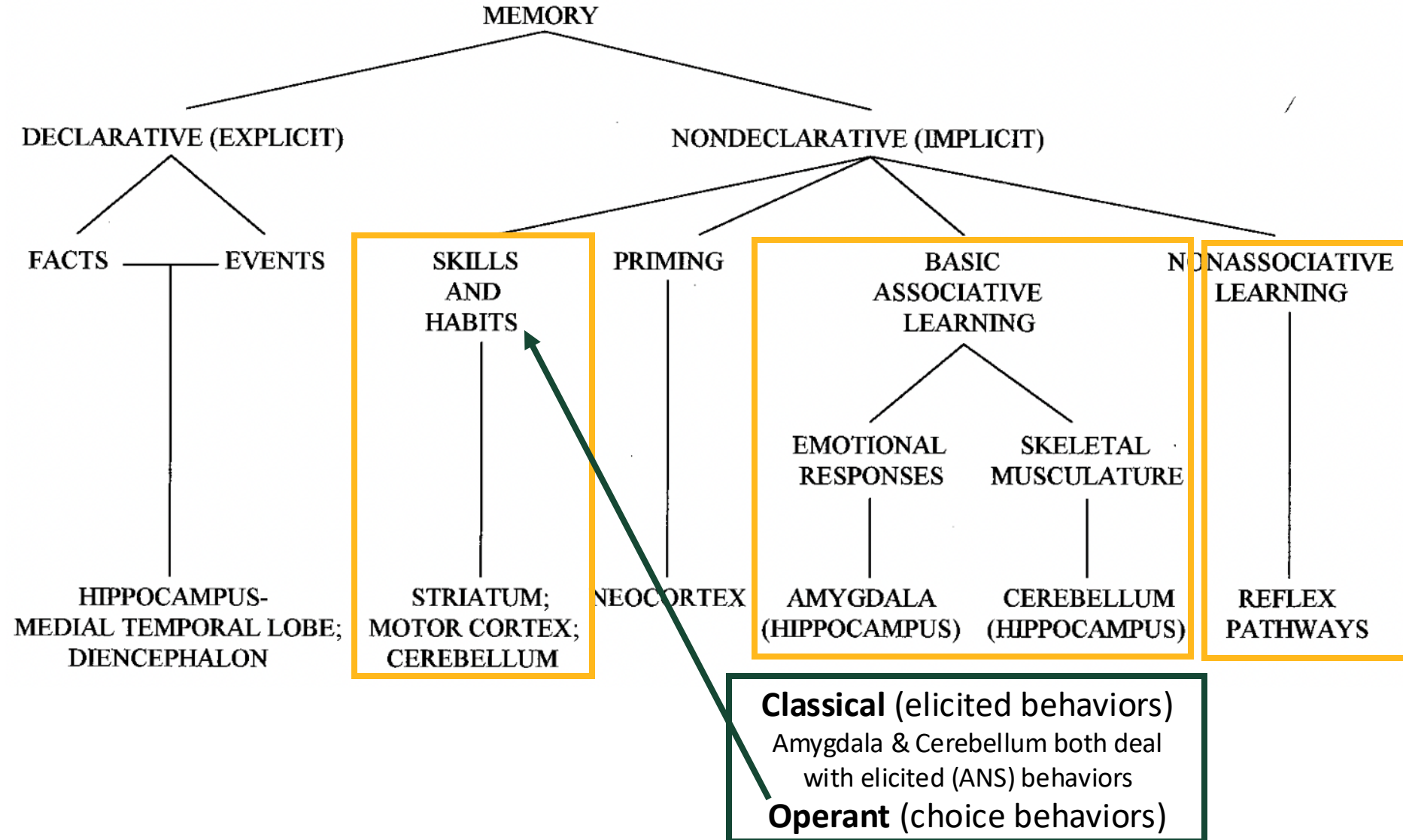


How do you think DA gets those neurons to wire together?



Representation of Environment ('S')





Questions?